

Arquivos do Projeto DEV Platform

Conteúdo da pasta: ./src/stake_file

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Índice

1. Arquivo: .env
2. Arquivo: .env.development
3. Arquivo: .env.production
4. Arquivo: .env.test
5. Arquivo: .gitignore
6. Arquivo: alembic.ini
7. Arquivo: composition_root.py
8. Arquivo: config.production.json
9. Arquivo: config.py
10. Arquivo: dtos.py
11. Arquivo: entities.py
12. Arquivo: exceptions.py
13. Arquivo: interfaces.py
14. Arquivo: logger.py
15. Arquivo: main.py
16. Arquivo: mappers.py
17. Arquivo: mkdocs.yml
18. Arquivo: models.py
19. Arquivo: mypy.ini
20. Arquivo: poetry.toml
21. Arquivo: ports.py
22. Arquivo: pyproject.toml
23. Arquivo: README.md
24. Arquivo: repositories.py
25. Arquivo: services.py
26. Arquivo: session.py
27. Arquivo: settings.json
28. Arquivo: structured_logger.py
29. Arquivo: tree.txt
30. Arquivo: unit_of_work.py
31. Arquivo: use_cases.py
32. Arquivo: user_commands.py
33. Arquivo: user_exceptions.py
34. Arquivo: validation_rules.py
35. Arquivo: value_objects.py

1. Arquivo: .env

```
# ./env
DB_USER_REMOTE="root"
DB_PASSWORD_REMOTE="Malato%2301"
DB_HOST_REMOTE="127.0.0.1:3306"
DB_NAME_REMOTE="env_management_db"

logging_level=DEBUG
```

2. Arquivo: .env.development

```
# ./env.development
# Nota 1: Este arquivo contém as variáveis de ambiente para o ambiente de desenvolvimento.
# Certifique-se de que este arquivo não seja incluído no controle de versão, pois contém informações sensíveis.
# Nota 2: Essas variáveis de ambiente estão em fase de transferência para melhor gestão de segredos em produção e para
# otimizar a validação de dados, visando maior segurança e eficiência.

# Configurações de ambiente para desenvolvimento
ENVIRONMENT=development
DATABASE_URL=mysql+aiomysql://root:Malato%2301@127.0.0.1:3306/user_management
DB_POOL_SIZE=5
DB_MAX_OVERFLOW=10
DB_ECHO=True
LOG_LEVEL=DEBUG
DEBUG=True

ALLOWED_DOMAINS="dominio1.com, dominio2.com"
VALIDATION_FORBIDDEN_WORDS="palavra1, palavra2"
ENTERPRISE_FORBIDDEN_WORDS="palavra3, palavra4"
ENTERPRISE_ALLOWED_DOMAINS="dominio3.com, dominio4.com"
```

3. Arquivo: .env.production

```
# ./env.production
# Nota 1: Este arquivo contém as variáveis de ambiente para o ambiente de produção.
# Certifique-se de que este arquivo não seja incluído no controle de versão, pois contém informações sensíveis.
# Nota 2 Essas variáveis de ambiente estão em fase de transferência para melhor gestão de segredos em produção e para
# otimizar a validação de dados, visando maior segurança e eficiência.

# Configurações de ambiente para produção
ENVIRONMENT=production
DATABASE_URL=mysql+aiomysql://root:Malato%2301@127.0.0.1:3306/user_management
DB_POOL_SIZE=20
DB_MAX_OVERFLOW=30
DB_ECHO=False
LOG_LEVEL=INFO
DEBUG=False

# Configurações de validação
VALIDATION_ENABLE_PROFANITY_FILTER=True
VALIDATION_FORBIDDEN_WORDS="palavrao1, palavrao2, termo_proibido" # Lista de palavras proibidas,
separadas por vírgula
VALIDATION_ALLOWED_DOMAINS="dominio1.com, dominio2.com" # Lista de domínios permitidos, separadas por vírgula
VALIDATION_BUSINESS_HOURS_ONLY=False

# Configurações de domínios permitidos para empresas
ENTERPRISE_ALLOWED_DOMAINS="empresa.com, company.com"

ALLOWED_DOMAINS=gmail.com,outrodominio.com
ENTERPRISE_FORBIDDEN_WORDS=
ENTERPRISE_ALLOWED_DOMAINS=
```

4. Arquivo: .env.test

```
# ./env.test
# Nota 1: Este arquivo contém as variáveis de ambiente para o ambiente de teste.
# Certifique-se de que este arquivo não seja incluído no controle de versão, pois contém informações sensíveis.
# Nota 2: Essas variáveis de ambiente estão em fase de transferência para melhor gestão de segredos em produção e para
# otimizar a validação de dados, visando maior segurança e eficiência.

# Configurações de ambiente para testes
ENVIRONMENT=test
DATABASE_URL=mysql+aiomysql://root:Malato%2301@127.0.0.1:3306/user_management
DB_POOL_SIZE=1
DB_MAX_OVERFLOW=0
DB_ECHO=False
LOG_LEVEL=WARNING
DEBUG=False

ALLOWED_DOMAINS="dominio1.com, dominio2.com"
VALIDATION_FORBIDDEN_WORDS="palavra1, palavra2"
ENTERPRISE_FORBIDDEN_WORDS="palavra3, palavra4"
ENTERPRISE_ALLOWED_DOMAINS="dominio3.com, dominio4.com"
```

5. Arquivo: .gitignore

```
# ./gitignore

# Dependências do sistema
node_modules/
chrome/

# Ambientes virtuais
.venv/
venv/
ENV/
env/

# Arquivos de cache Python
__pycache__/
*.py[cod]
*$py.class
.pytest_cache/
.coverage
htmlcov/
.tox/
.nox/

# Distribuição / empacotamento
dist/
build/
*.egg-info/
*.egg

# Documentação gerada pelo MkDocs
site/

# Poetry
poetry.lock
# NOTA:
# 1 - Descomente a linha acima se NÃO quiser versionar o poetry.lock
# 2 - Commit do arquivo poetry.lock permite criar um ambiente determinístico

# Arquivos de log
*.log
logs/

# Arquivos temporários
*.tmp
*.bak
*.swp
*~

# Arquivos do sistema operacional
.DS_Store
Thumbs.db

# IDEs e editores
.idea/
.vscode/
*.sublime-project
*.sublime-workspace
.project
.pydevproject
.spyderproject
.spyproject
.ropeproject

# Jupyter Notebook
```

.ipynb_checkpoints

Arquivos de ambiente

.env

.env.local

.env.development.local

.env.test.local

.env.production.local

.env.development

.env.test

.env.production

.env*

Diretórios de mídia/uploads (se aplicável)

media/

uploads/

SQLite DB (se aplicável)

*.sqlite3

*.db

Arquivos de configuração local

local_settings.py

6. Arquivo: alembic.ini

```
# ./alembic.ini
# A generic, single database configuration.

[alembic]
# path to migration scripts
# Use forward slashes (/) also on windows to provide an os agnostic path
script_location = migrations

# template used to generate migration file names; The default value is %(rev)s_%(slug)s
# Uncomment the line below if you want the files to be prepended with date and time
# see https://alembic.sqlalchemy.org/en/latest/tutorial.html#editing-the-ini-file
# for all available tokens
# file_template = %(year)d_%(month).2d_%(day).2d_%(hour).2d_%(minute).2d-%(rev)s_%(slug)s

# sys.path path, will be prepended to sys.path if present.
# defaults to the current working directory.
prepend_sys_path = .

# timezone to use when rendering the date within the migration file
# as well as the filename.
# If specified, requires the python>=3.9 or backports.zoneinfo library and tzdata library.
# Any required deps can installed by adding `alembic[tz]` to the pip requirements
# string value is passed to ZoneInfo()
# leave blank for localtime
# timezone =

# max length of characters to apply to the "slug" field
# truncate_slug_length = 40

# set to 'true' to run the environment during
# the 'revision' command, regardless of autogenerate
# revision_environment = false

# set to 'true' to allow .pyc and .pyo files without
# a source .py file to be detected as revisions in the
# versions/ directory
# sourceless = false

# version location specification; This defaults
# to migrations/versions. When using multiple version
# directories, initial revisions must be specified with --version-path.
# The path separator used here should be the separator specified by "version_path_separator" below.
# version_locations = %(here)s/bar:%(here)s/bat:migrations/versions

# version path separator; As mentioned above, this is the character used to split
# version_locations. The default within new alembic.ini files is "os", which uses os.pathsep.
# If this key is omitted entirely, it falls back to the legacy behavior of splitting on spaces and/or commas.
# Valid values for version_path_separator are:
#
# version_path_separator = :
# version_path_separator = ;
# version_path_separator = space
# version_path_separator = newline
#
# Use os.pathsep. Default configuration used for new projects.
version_path_separator = os

# set to 'true' to search source files recursively
# in each "version_locations" directory
# new in Alembic version 1.10
# recursive_version_locations = false
```

```

# the output encoding used when revision files
# are written from script.py.mako
# output_encoding = utf-8

sqlalchemy.url = "mysql+aiomysql://root:Malato#01@127.0.0.1:3306/user_management" # driver://user:pass@localhost/dbname

[post_write_hooks]
# post_write_hooks defines scripts or Python functions that are run
# on newly generated revision scripts. See the documentation for further
# detail and examples

# format using "black" - use the console_scripts runner, against the "black" entrypoint
# hooks = black
# black.type = console_scripts
# black.entrypoint = black
# black.options = -l 79 REVISION_SCRIPT_FILENAME

# lint with attempts to fix using "ruff" - use the exec runner, execute a binary
# hooks = ruff
# ruff.type = exec
# ruff.executable = %(here)s/.venv/bin/ruff
# ruff.options = check --fix REVISION_SCRIPT_FILENAME

# Logging configuration
[loggers]
keys = root,sqlalchemy,alembic

[handlers]
keys = console

[formatters]
keys = generic

[logger_root]
level = WARNING
handlers = console
qualname =

[logger_sqlalchemy]
level = WARNING
handlers =
qualname = sqlalchemy.engine

[logger_alembic]
level = INFO
handlers =
qualname = alembic

[handler_console]
class = StreamHandler
args = (sys.stderr,)
level = NOTSET
formatter = generic

[formatter_generic]
format = %(levelname)-5.5s [%(name)s] %(message)s
datefmt = %H:%M:%S

```

7. Arquivo: composition_root.py

```
# ./src/dev_platform/infrastructure/composition_root.py
# -*- coding: utf-8 -*-
"""
Este módulo define a raiz de composição para injeção de dependências,
centralizando a criação e configuração de todas as dependências da aplicação.
Agora utiliza um ValidationRuleProvider para desacoplar a lógica de regras de validação,
respeitando OCP e separando logging de infraestrutura.
"""

from typing import Any, Dict, List, Optional
from dev_platform.application.user.use_cases import (
    CreateUserUseCase,
    ListUsersUseCase,
    UpdateUserUseCase,
    GetUserUseCase,
    DeleteUserUseCase,
)
from dev_platform.domain.user.services import (
    UserDomainService,
    UserAnalyticsService,
    UserUniquenessService,
    UserValidatorService
)
from dev_platform.domain.validation_rules import (
    EmailFormatAdvancedValidationRule,
    NameContentValidationRule,
    EmailDomainValidationRule,
    BusinessHoursValidationRule,
    NameProfanityValidationRule,
    ForbiddenWordsValidationRule
)

from dev_platform.infrastructure.logging.structured_logger import StructuredLogger
from dev_platform.infrastructure.config import ConfigurationFacade
from dev_platform.application.ports.logger import ILogger
from dev_platform.domain.validation_rules import ValidationRule

from dev_platform.application.user.ports import UnitOfWork # Importa a interface
from dev_platform.domain.user.interfaces import IUserRepository # Importa a interface
from dev_platform.infrastructure.database.unit_of_work import SQLUnitOfWork # Ainda precisa da imple
mentação concreta para instanciar
from dev_platform.infrastructure.database.repositories import SQLUserRepository # Ainda precisa da impl
mentação concreta para instanciar

class ValidationRuleProvider:
    """Provider para regras de validação, desacoplado da infraestrutura."""
    def __init__(
        self,
        default_allowed_domains: List[str],
        default_forbidden_words: List[str],
        enterprise_allowed_domains: List[str],
        enterprise_forbidden_words: List[str],
        enable_profanity_filter: bool,
        logger: ILogger
    ):
        self.default_allowed_domains = default_allowed_domains
        self.default_forbidden_words = default_forbidden_words
        self.enterprise_allowed_domains = enterprise_allowed_domains
        self.enterprise_forbidden_words = enterprise_forbidden_words
        self.enable_profanity_filter = enable_profanity_filter
        self.logger = logger
```

```

def _default_rules(self) -> List[ValidationRule]:
    """
    Regras de validação padrão para usuários comuns.
    Essas regras são aplicadas a todos os usuários, exceto os enterprise.
    """
    return [
        EmailFormatAdvancedValidationRule(),
        NameContentValidationRule(),
        EmailDomainValidationRule(self._default_allowed_domains),
        NameProfanityValidationRule(forbidden_words=self._default_forbidden_words)
    ]

def get_rules(self, user_type: str = "default") -> List[ValidationRule]:
    """
    Retorna a lista de regras de validação conforme o tipo de usuário.
    """
    if user_type == "enterprise":
        return self._enterprise_rules()
    return self._default_rules()

def _enterprise_rules(self) -> List[ValidationRule]:
    """
    Regras de validação específicas para usuários enterprise.
    Permite maior controle e customização para clientes corporativos.
    """
    if self._enable_profanity_filter and not self._enterprise_forbidden_words:
        self._logger.warning("Profanity filter enabled for enterprise, but forbidden words list is empty.")

    rules: List[ValidationRule] = [
        EmailFormatAdvancedValidationRule(),
        NameContentValidationRule(),
        EmailDomainValidationRule(self._enterprise_allowed_domains),
        NameProfanityValidationRule(forbidden_words=self._enterprise_forbidden_words),
        # Regras extras para enterprise:
        ForbiddenWordsValidationRule(forbidden_words=self._enterprise_forbidden_words),
        BusinessHoursValidationRule(True), # Exemplo: só permitir operações em horário comercial
    ]
    return rules

class CompositionRoot:
    """
    Composition root for dependency injection.
    Centraliza a criação e configuração de todas as dependências da aplicação.
    Utiliza ValidationRuleProvider para regras de validação (OCP).
    """

    def __init__(
        self,
        config: ConfigurationFacade, # Configuração injetada
        logger: ILogger, # Logger injetado
    ):
        self._config = config
        self._logger = logger
        # A CompositionRoot solicita os dados já formatados para a facade de configuração.
        self._validation_rule_provider = ValidationRuleProvider(
            default_allowed_domains=self._config.get_list("allowed_domains"),
            default_forbidden_words=self._config.get_list("validation_forbidden_words"),
            enterprise_allowed_domains=self._config.get_list("enterprise_allowed_domains"),
            enterprise_forbidden_words=self._config.get_list("enterprise_forbidden_words"),
            enable_profanity_filter=self._config.get_typed("validation_enable_profanity_filter", False, bool),
            logger=self._logger
        )

    def create_user_use_case(self, uow: UnitOfWork, user_repository: IUserRepository) -> CreateUserUseCase:

```

```

# O domain_service deve receber apenas o que ele precisa para as regras de domínio.
return CreateUserUseCase(
    uow=uow,
    domain_service=self.user_domain_service(),
    user_uniqueness_service=self.user_uniqueness_service(user_repository),
    logger=self._logger,
)

# Novo método para criar o UoW, que encapsula a criação do repositório concreto
def create_unit_of_work(self) -> UnitOfWork:
    # O SQLUnitOfWork agora recebe o repositório via injeção ou uma factory
    # Para simplificar, vamos injetar o logger e o repositório aqui
    user_repo = SQLUserRepository(session=None, logger=self._logger) # A sessão será injetada pelo U
oW
    return SQLUnitOfWork(logger=self._logger, user_repository=user_repo) # Passa o repositório concre
to

def list_users_use_case(self, uow: SQLUnitOfWork, user_repository: IUserRepository) -> ListUsersUseCas
e:
    return ListUsersUseCase(
        uow=uow,
        logger=self._logger
    )

def update_user_use_case(self, uow: SQLUnitOfWork, user_repository: IUserRepository) -> UpdateUser
UseCase:
    return UpdateUserUseCase(
        uow=uow,
        domain_service=self.user_domain_service(user_repository),
        logger=self._logger,
    )

def get_user_use_case(self, uow: SQLUnitOfWork, user_repository: IUserRepository) -> GetUserUseCase:
    return GetUserUseCase(
        uow=uow,
        domain_service=self.user_domain_service(user_repository),
        logger=self._logger,
    )

def delete_user_use_case(self, uow: SQLUnitOfWork, user_repository: IUserRepository) -> DeleteUserUs
eCase:
    return DeleteUserUseCase(
        uow=uow,
        domain_service=self.user_domain_service(user_repository),
        logger=self._logger,
    )

def user_domain_service(self, user_type: str = "default") -> UserValidatorService:
    """
    Cria UserValidatorService com regras de validação baseadas em configuração e tipo de usuário.
    """
    rules = self._validation_rule_provider.get_rules(user_type)
    return UserValidatorService(validation_rules=rules)

def user_uniqueness_service(self, user_repository: IUserRepository) -> UserUniquenessService:
    """
    Cria UserUniquenessService, que depende do repositório.
    """
    return UserUniquenessService(user_repository)

def user_analytics_service(self, user_repository: IUserRepository) -> UserAnalyticsService:
    """
    Cria o serviço de analytics de usuários.
    """
    return UserAnalyticsService(user_repository)

```

```
def create_enterprise_user_domain_service(  
    self, user_repository: IUserRepository  
) -> UserDomainService:  
    return self.user_domain_service(user_repository, user_type="enterprise")
```

8. Arquivo: config.production.json

```
{
  "name": "dev_platform",
  "version": "1.0.0",
  "description": "Production configuration for the development platform",
  "environment": "production",
  "apiBaseUrl": "https://api.devplatform.com",
  "loggingLevel": "error",
  "features": {
    "enableFeatureX": false,
    "enableFeatureY": true
  },
  "database": {
    "databasename": "user_management",
    "host": "127.0.0.1",
    "port": 3306,
    "username": "root",
    "password": "Malato%2301"
  }
}
```

9. Arquivo: config.py

```
# ./src/dev_platform/infrastructure/config.py
# -*- coding: utf-8 -*-
"""
Este módulo define a infraestrutura de configuração do DEV Platform,
responsável por carregar e acessar as configurações do sistema.
"""

import os
import json
from typing import List, Dict, Any, Optional, Callable, Type
from dotenv import load_dotenv
from dev_platform.domain.exceptions import ConfigurationException
from dev_platform.application.ports.logger import ILogger

class EnvLoader:
    """
    Responsável por carregar variáveis de ambiente de arquivos .env.
    """
    def __init__(self, environment: str, logger: ILogger):
        self.environment = environment
        self._logger: ILogger = logger

    def load(self) -> None:
        """
        Carrega variáveis de ambiente do arquivo .env.<environment>.
        """
        dotenv_path = f".env.{self.environment}"
        base_dir = os.path.abspath(os.path.join(os.path.dirname(__file__), "..", "..", ".."))
        full_dotenv_path = os.path.join(base_dir, dotenv_path)
        if os.path.exists(full_dotenv_path):
            load_dotenv(dotenv_path=full_dotenv_path, override=True)
        else:
            if self.environment == "production":
                self._logger.info(
                    f"Arquivo .env.{self.environment} não encontrado em {full_dotenv_path}. Assumindo que as v"
                    "ariáveis de ambiente são configuradas externamente."
                )
            else:
                self._logger.info(
                    f"AVISO: Arquivo .env.{self.environment} não encontrado em {full_dotenv_path}. Algumas vari"
                    "áveis de ambiente podem não estar definidas."
                )

class JsonConfigLoader:
    """
    Responsável por carregar configurações de arquivos JSON.
    """
    def __init__(self, environment: str, logger: ILogger):
        self.environment = environment
        self._logger: ILogger = logger

    def load(self) -> Dict[str, Any]:
        """
        Carrega configurações do arquivo config.<environment>.json.
        """
        config_file_path = f"config.{self.environment}.json"
        base_dir = os.path.abspath(os.path.join(os.path.dirname(__file__), "..", "..", ".."))
        full_config_file_path = os.path.join(base_dir, config_file_path)
        config: Dict[str, Any] = {}
        if os.path.exists(full_config_file_path):
            try:
```



```

        with open(full_config_file_path, "r") as f:
            config = json.load(f)
        except Exception as e:
            self._logger.error(
                "Erro ao carregar o arquivo de configuração.",
                config_key="CONFIG_FILE_LOAD_ERROR"
            )
            raise ConfigurationException(
                config_key="CONFIG_FILE_LOAD_ERROR",
                reason=f"Erro ao carregar o arquivo de configuração {full_config_file_path}: {e}"
            )
        else:
            self._logger.info(
                f"Arquivo de {full_config_file_path} não encontrado. Usando apenas variáveis de ambiente e padrões.",
                config_key="CONFIG_FILE_NOT_FOUND"
            )
        return config

```

class ConfigValidator:

```

    """
    Responsável por validar configurações críticas.
    """

    def __init__(self, environment: str, logger: ILogger):
        self.environment = environment
        self._logger: ILogger = logger

    def validate(self) -> None:
        """
        Valida se as configurações críticas estão presentes (ex: DATABASE_URL em produção).
        """
        if self.environment == "production":
            if not os.getenv("DATABASE_URL"):
                self._logger.error(
                    "DATABASE_URL não configurada para produção.",
                    config_key="DATABASE_URL"
                )
            raise ConfigurationException(
                config_key="DATABASE_URL",
                reason="DATABASE_URL must be set in production environment."
            )

```

class DatabaseDriverChecker:

```

    """
    Responsável por garantir o uso de drivers assíncronos na URL do banco de dados.
    """

    @staticmethod
    def ensure_async_driver(url: str) -> str:
        """
        Ajusta a URL do banco para usar driver assíncrono, se necessário.
        """
        if url.startswith("mysql://"):
            return url.replace("mysql://", "mysql+aiomysql://")
        elif url.startswith("postgresql://"):
            return url.replace("postgresql://", "postgresql+asyncpg://")
        elif url.startswith("sqlite://"):
            return url.replace("sqlite://", "sqlite+aiosqlite://")
        return url

```

class ConfigAccessor:

```

    """
    Responsável por acessar valores de configuração.
    """

```

```

def __init__(self, config: Dict[str, Any], logger: ILogger):
    self._config = config
    self._logger: ILogger = logger

def get(self, key: str, default: Any = None) -> Any:
    """
    Obtém um valor de configuração, preferindo variáveis de ambiente.
    """
    env_key = key.upper().replace(".", "_")
    env_value = os.getenv(env_key)
    if env_value is not None:
        return env_value
    value = self._config.get(key, default)
    if value is None:
        self._logger.error(
            f"Configuração obrigatória '{key}' não encontrada.",
            config_key=key
        )
        raise ConfigurationException(
            config_key=key,
            reason=f"Required configuration '{key}' is missing."
        )
    return value

def get_typed(self, key: str, default: Any = None, cast_type: Type = str) -> Any:
    """
    Obtém um valor de configuração e converte para o tipo desejado.
    """
    value = self.get(key, default)
    if value is None:
        return default
    try:
        if cast_type is bool:
            return str(value).strip().lower() in ("1", "true", "yes", "on")
        return cast_type(value)
    except Exception:
        return default

def get_list(self, key: str, default: Optional[List[str]] = None) -> List[str]:
    """Obtém um valor de configuração como uma lista de strings a partir de um valor separado por vírgulas."""
    value = self.get(key, None) # Usando get() para pegar o valor bruto
    if value is None:
        return default if default is not None else []
    if isinstance(value, list):
        return value
    if not isinstance(value, str) or not value.strip():
        return []
    return [item.strip() for item in value.split(',') if item.strip()]

def get_all_config(self) -> Dict[str, Any]:
    """
    Retorna todas as configurações efetivas, mesclando arquivo JSON e variáveis de ambiente.
    """
    all_configs = self._config.copy()
    for env_key, env_value in os.environ.items():
        all_configs[env_key.lower().replace(".", "_")] = env_value
    return all_configs

class ConfigurationFacade:
    """
    Fachada para acesso às configurações do sistema, permitindo injeção de dependências.

    Parâmetros de __init__:
    logger: ILogger customizado (opcional)
    """

```

```

env_loader_factory: Callable para criar um EnvLoader customizado (opcional)
json_loader_factory: Callable para criar um JsonConfigLoader customizado (opcional)
validator_factory: Callable para criar um ConfigValidator customizado (opcional)
accessor_factory: Callable para criar um ConfigAccessor customizado (opcional)
"""

def __init__(
    self,
    logger: Optional[ILogger] = None,
    env_loader_factory: Optional[Callable[[str, ILogger], EnvLoader]] = None,
    json_loader_factory: Optional[Callable[[str, ILogger], JsonConfigLoader]] = None,
    validator_factory: Optional[Callable[[str, ILogger], ConfigValidator]] = None,
    accessor_factory: Optional[Callable[[Dict[str, Any], ILogger], ConfigAccessor]] = None,
    environment: Optional[str] = None,
) -> None:
    """
    Inicializa a fachada de configuração, permitindo injeção de dependências para testes ou customizaçã
    o.
    """
    if hasattr(self, "_initialized") and self._initialized:
        return
    self._environment: str = environment or os.getenv("ENVIRONMENT", "production")

    if logger is not None:
        self._logger: ILogger = logger
    else:
        from dev_platform.infrastructure.logging.structured_logger import StructuredLogger
        self._logger: ILogger = StructuredLogger()

    # Permite injeção de dependências para facilitar testes/mocks
    env_loader: EnvLoader = (env_loader_factory or EnvLoader)(self._environment, self._logger)
    env_loader.load()
    json_loader: JsonConfigLoader = (json_loader_factory or JsonConfigLoader)(self._environment, self._l
ogger)
    config_dict: Dict[str, Any] = json_loader.load()
    validator: ConfigValidator = (validator_factory or ConfigValidator)(self._environment, self._logger)
    validator.validate()
    self._accessor: ConfigAccessor = (accessor_factory or ConfigAccessor)(config_dict, self._logger)
    self._initialized: bool = True

    @classmethod
    def _reset_singleton(cls) -> None:
        """
        Reseta a instância singleton (apenas para uso em testes).
        """
        cls._instance = None

    def get(self, key: str, default: Any = None) -> Any:
        """
        Obtém um valor de configuração, preferindo variáveis de ambiente.
        """
        return self._accessor.get(key, default)

    def get_typed(self, key: str, default: Any = None, cast_type: Type = str) -> Any:
        """
        Obtém um valor de configuração e converte para o tipo desejado.
        """
        return self._accessor.get_typed(key, default, cast_type)

    def get_list(self, key: str, default: Optional[List[str]] = None) -> List[str]:
        """Obtém um valor de configuração como uma lista de strings a partir de um valor separado por vír
gulas."""
        value = self.get(key, None)
        if value is None:
            return default if default is not None else []
        if isinstance(value, list):

```

```

        return value
    if not isinstance(value, str) or not value.strip():
        return []
    return [item.strip() for item in value.split(',') if item.strip()]
    # return self._accessor.get_list(key, default)

```

```
def get_all_config(self) -> Dict[str, Any]:
```

```

    """
    Retorna todas as configurações efetivas, mesclando arquivo JSON e variáveis de ambiente.
    """
    return self._accessor.get_all_config()

```

```
@property
```

```
def database_url(self) -> str:
```

```

    """
    Retorna a URL do banco de dados com driver assíncrono garantido.
    """
    url = self.get("DATABASE_URL")
    if not url:
        raise ConfigurationException(
            config_key="DATABASE_URL",
            reason="DATABASE_URL is not configured for the current environment."
        )
    return DatabaseDriverChecker.ensure_async_driver(url)

```

```
@property
```

```
def sync_database_url(self) -> str:
```

```

    """
    Retorna a URL do banco de dados sem garantir driver assíncrono.
    """
    url = self.get("DATABASE_URL")
    if not url:
        raise ConfigurationException(
            config_key="DATABASE_URL",
            reason="DATABASE_URL is not configured for the current environment."
        )
    return url

```

10. Arquivo: dtos.py

```
# ./src/dev_platform/application/user/dtos.py
# -*- coding: utf-8 -*-
"""
Este módulo define os Data Transfer Objects (DTOs) para a entidade User,
permitindo a transferência de dados entre camadas da aplicação.
"""

from typing import Optional
from pydantic import BaseModel, StrictStr, EmailStr, field_validator

class UserDTO(BaseModel):
    """
    Data Transfer Object for User entity.
    """
    id: StrictStr
    name: StrictStr
    email: StrictStr

    class Config:
        """
        Configurações do Pydantic para o DTO.
        """
        from_attributes = True

class UserCreatedDTO(BaseModel):
    """
    Data Transfer Object for creating a new User.
    """
    name: Optional[StrictStr] = None
    email: Optional[EmailStr] = None

    @field_validator("name", mode="before")
    def validate_name(cls, v):
        """
        Valida o nome do usuário.
        """
        if not v or len(v) == 0:
            raise ValueError("Precisa ser um nome, o campo não pode ficar vazio")
        return v.strip()

    @field_validator("email", mode="before")
    def validate_email(cls, v):
        """
        Valida o e-mail do usuário.
        """
        if not v or len(v) == 0:
            raise ValueError("Precisa ser um e-mail")
        return v.lower().strip()

    class Config:
        """
        Configurações do Pydantic para o DTO de criação.
        """
        from_attributes = True

class UserUpdateDTO(BaseModel):
    """
    Data Transfer Object for updating an existing User.
    """
    name: Optional[StrictStr] = None
    email: Optional[EmailStr] = None
```

```
@field_validator("name", mode="before")
def validate_name(cls, v):
    """
    Valida o nome do usuário para atualização.
    """
    return v.strip()

@field_validator("email", mode="before")
def validate_email(cls, v):
    """
    Valida o e-mail do usuário para atualização.
    """
    return v.lower().strip()

class Config:
    """
    Configurações do Pydantic para o DTO de atualização.
    """
    from_attributes = True
```

11. Arquivo: entities.py

```
# ./src/dev_platform/domain/user/entities.py
# -*- coding: utf-8 -*-
"""
Este módulo define a entidade User, representando um usuário do sistema.
A entidade é imutável e utiliza Value Objects para validação de nome e e-mail.
"""

from dataclasses import dataclass, replace
from typing import Optional
from dev_platform.domain.user.value_objects import Email, UserName

@dataclass(frozen=True)
class User:
    """Entidade de domínio representando um usuário (imutável)."""
    id: Optional[int]
    name: UserName
    email: Email

    @classmethod
    def create(cls, name: str, email: str) -> "User":
        """Cria um novo usuário, validando nome e e-mail via Value Objects."""
        return cls(id=None, name=UserName(name), email=Email(email))

    def with_id(self, new_id: int) -> "User":
        """Retorna uma nova instância do usuário com o id atualizado."""
        return replace(self, id=new_id)

    def update_details(self, new_name: str, new_email: str) -> "User":
        """Retorna uma nova instância do usuário com nome e e-mail atualizados."""
        return replace(self, name=UserName(new_name), email=Email(new_email))
```

12. Arquivo: exceptions.py

```
# ./src/dev_platform/domain/exceptions.py
# -*- coding: utf-8 -*-
"""
Este módulo define exceções para a camada de aplicação e infraestrutura,
permitindo o tratamento de erros de forma consistente e informativa.
"""

from datetime import datetime
from typing import Optional, Dict, Any

# Application layer exceptions
class ApplicationException(Exception):
    """Base exception for application layer errors."""

    def __init__(self, message: str, original_exception: Optional[Exception] = None):
        self.message = message
        self.original_exception = original_exception
        self.timestamp = datetime.now()
        super().__init__(self.message)

    def __str__(self):
        return f"[ApplicationException] {self.message}"

class UseCaseException(ApplicationException):
    """Raised when a use case execution fails."""

    def __init__(
        self,
        use_case_name: str,
        reason: str,
        original_exception: Optional[Exception] = None,
    ):
        self.use_case_name = use_case_name
        self.reason = reason
        super().__init__(
            message=f"Use case '{use_case_name}' failed: {reason}",
            original_exception=original_exception,
        )

    def __str__(self):
        return f"[UseCaseException] {self.message}"

# Infrastructure layer exceptions
class InfrastructureException(Exception):
    """Base exception for infrastructure layer errors."""

    def __init__(
        self,
        message: str,
        component: str,
        original_exception: Optional[Exception] = None,
    ):
        self.message = message
        self.component = component
        self.original_exception = original_exception
        self.timestamp = datetime.now()
        super().__init__(self.message)

    def to_dict(self) -> Dict[str, Any]:
        """Convert exception to dictionary for logging/serialization."""
        return {
            "message": self.message,
            "component": self.component,
```



```

        "timestamp": self.timestamp.isoformat(),
        "original_error": str(self.original_exception)
        if self.original_exception
        else None,
    }

    def __str__(self):
        return f"[InfrastructureException:{self.component}] {self.message}"

class DatabaseException(InfrastructureException):
    """Raised when database operations fail."""

    def __init__(
        self,
        operation: str,
        reason: str,
        original_exception: Optional[Exception] = None,
    ):
        self.operation = operation
        self.reason = reason
        super().__init__(
            message=f"Database operation '{operation}' failed: {reason}",
            component="database",
            original_exception=original_exception,
        )

    def to_dict(self) -> Dict[str, Any]:
        """Extended dictionary representation for database errors."""
        base_dict = super().to_dict()
        base_dict.update({"operation": self.operation, "reason": self.reason})
        return base_dict

    def __str__(self):
        return f"[DatabaseException:{self.operation}] {self.message}"

class ConfigurationException(InfrastructureException):
    """Raised when configuration is invalid or missing."""

    def __init__(self, config_key: str, reason: str):
        self.config_key = config_key
        self.reason = reason
        super().__init__(
            message=f"Configuration error for '{config_key}': {reason}",
            component="configuration",
        )

    def __str__(self):
        return f"[ConfigurationException:{self.config_key}] {self.message}"

class CacheException(InfrastructureException):
    """Raised when cache operations fail."""

    def __init__(
        self,
        operation: str,
        key: str,
        reason: str,
        original_exception: Optional[Exception] = None,
    ):
        self.operation = operation
        self.key = key
        self.reason = reason
        super().__init__(
            message=f"Cache {operation} failed for key '{key}': {reason}",
            component="cache",
            original_exception=original_exception,
        )

```

```

    )

    def __str__(self):
        return f"[CacheException:{self.operation}] {self.message}"

# Repository-specific exceptions
class RepositoryException(InfrastructureException):
    """Base exception for repository layer errors."""

    def __init__(
        self,
        repository_name: str,
        operation: str,
        reason: str,
        original_exception: Optional[Exception] = None,
    ):
        self.repository_name = repository_name
        self.operation = operation
        self.reason = reason
        super().__init__(
            message=f"Repository '{repository_name}' {operation} failed: {reason}",
            component="repository",
            original_exception=original_exception,
        )

    def __str__(self):
        return f"[RepositoryException:{self.repository_name}.{self.operation}] {self.message}"

class DataIntegrityException(RepositoryException):
    """Raised when data integrity constraints are violated."""

    def __init__(
        self,
        constraint_name: str,
        details: str,
        original_exception: Optional[Exception] = None,
    ):
        self.constraint_name = constraint_name
        self.details = details
        super().__init__(
            repository_name="database",
            operation="constraint_validation",
            reason=f"Constraint '{constraint_name}' violated: {details}",
            original_exception=original_exception,
        )

    def __str__(self):
        return f"[DataIntegrityException:{self.constraint_name}] {self.message}"

class DataCorruptionException(RepositoryException):
    """Raised when data corruption is detected."""

    def __init__(self, entity_type: str, entity_id: str, corruption_details: str):
        self.entity_type = entity_type
        self.entity_id = entity_id
        self.corruption_details = corruption_details
        super().__init__(
            repository_name="database",
            operation="data_validation",
            reason=f"{entity_type} {entity_id} has corrupted data: {corruption_details}",
        )

    def __str__(self):
        return f"[DataCorruptionException:{self.entity_type}:{self.entity_id}] {self.message}"

```

13. Arquivo: interfaces.py

```
#!/src/dev_platform/domain/user/interfaces.py
# -*- coding: utf-8 -*-
"""
Este módulo define a interface IUserRepository, que especifica os métodos necessários
para manipulação de usuários no repositório.
"""

from abc import ABC, abstractmethod
from typing import List, Optional
from dev_platform.domain.user.entities import User
from dev_platform.domain.user.value_objects import Email

class IUserRepository(ABC):
    @abstractmethod
    async def add(self, user: User) -> User:
        """Adiciona um novo usuário ao repositório."""
        pass

    @abstractmethod
    async def update(self, user: User) -> User:
        """Atualiza um usuário existente no repositório."""
        pass

    @abstractmethod
    async def find_by_email(self, email: Email) -> Optional[User]:
        """Busca um usuário pelo e-mail. Retorna o usuário ou None se não encontrado."""
        pass

    @abstractmethod
    async def find_all(self) -> List[User]:
        """Busca todos os usuários."""
        pass

    @abstractmethod
    async def find_by_id(self, user_id: int) -> Optional[User]:
        """Busca um usuário pelo ID. Retorna o usuário ou None se não encontrado."""
        pass

    @abstractmethod
    async def delete(self, user_id: int) -> bool:
        """Deleta um usuário pelo ID."""
        pass

    @abstractmethod
    async def find_by_name_contains(self, name_part: str) -> List[User]:
        """Busca usuários por uma parte do nome."""
        pass

    @abstractmethod
    async def count(self) -> int:
        """Conta o número total de usuários."""
        pass
```

14. Arquivo: logger.py

```
# src/dev_platform/application/ports/logger.py
# -*- coding: utf-8 -*-
"""
Este módulo define a interface ILogger para um logger estruturado,
permitindo a injeção de dependências e abstraindo a implementação específica de logging.
"""

from abc import ABC, abstractmethod
from typing import Optional

class ILogger(ABC):
    """Interface for a structured logger."""

    @abstractmethod
    def set_correlation_id(self, correlation_id: Optional[str] = None):
        """Set a correlation ID for tracking logs across requests."""
        raise NotImplementedError

    @abstractmethod
    def info(self, message: str, **kwargs):
        """Log an informational message."""
        raise NotImplementedError

    @abstractmethod
    def error(self, message: str, **kwargs):
        """Log an error message."""
        raise NotImplementedError

    @abstractmethod
    def warning(self, message: str, **kwargs):
        """Log a warning message."""
        raise NotImplementedError

    @abstractmethod
    def debug(self, message: str, **kwargs):
        """Log a debug message."""
        raise NotImplementedError

    @abstractmethod
    def critical(self, message: str, **kwargs):
        """Log a critical message."""
        raise NotImplementedError
```

15. Arquivo: main.py

```
# ./src/dev_platform/main.py
# -*- coding: utf-8 -*-
"""
Este módulo define a interface de linha de comando (CLI) para o DEV Platform,
permitindo a interação com os comandos relacionados a usuários.
"""

import click
# Importe user_cli do user_commands (renomeado para evitar conflito)
from dev_platform.interface.cli.user_commands import cli

# Cria um grupo Click principal
@click.group()
def main_cli():
    """CLI para o DEV Platform."""
    pass

# Adiciona os comandos de usuário como um subgrupo 'user'
main_cli.add_command(cli, name="user")

if __name__ == "__main__":
    main_cli()
```

16. Arquivo: mappers.py

```
# ./src/dev_platform/application/user/mappers.py
# -*- coding: utf-8 -*-
"""
Este módulo define os mapeadores para converter entre entidades User e seus Data Transfer Objects (DTOs),
permitindo a transferência de dados entre camadas da aplicação.
"""

from dev_platform.domain.user.entities import User
from dev_platform.application.user.dtos import UserDTO, UserCreateDTO, UserUpdateDTO
from typing import List

def user_to_dto(user: User) -> UserDTO:
    return UserDTO(id=str(user.id), name=user.name.value, email=user.email.value)

def dto_to_user(dto: UserDTO) -> User:
    return User.create(name=dto.name, email=dto.email)

def create_dto_to_user(dto: UserCreateDTO) -> User:
    return User.create(name=dto.name, email=dto.email)

def update_dto_to_user(user_id: str, dto: UserUpdateDTO) -> User:
    # Supondo que User.create aceita id como argumento opcional
    return User.create(id=user_id, name=dto.name, email=dto.email)

def users_to_dtos(users: List[User]) -> List[UserDTO]:
    return [user_to_dto(user) for user in users]

# Para get, basta usar user_to_dto
# Para delete, normalmente só o id é necessário, não precisa de conversão extra
```

17. Arquivo: mkdocs.yml

```
site_name: Dev Platform
site_description: Descrição do seu projeto
site_author: Seu Nome
site_url: https://exemplo.com/

repo_name: seu-usuario/seu-projeto
repo_url: https://github.com/seu-usuario/seu-projeto

theme:
  name: material
  features:
    - navigation.instant
    - navigation.tracking
    - navigation.expand
    - navigation.indexes
    - navigation.top
    - search.highlight
    - search.share
  palette:
    - media: "(prefers-color-scheme: light)"
      scheme: default
      primary: indigo
      accent: indigo
      toggle:
        icon: material/toggle-switch-off-outline
        name: Mudar para modo escuro
    - media: "(prefers-color-scheme: dark)"
      scheme: slate
      primary: indigo
      accent: indigo
      toggle:
        icon: material/toggle-switch
        name: Mudar para modo claro

markdown_extensions:
  - admonition
  - attr_list
  - def_list
  - footnotes
  - md_in_html
  - pymdownx.details
  - pymdownx.emoji:
      emoji_index: !!python/name:material.extensions.emoji.twemoji
      emoji_generator: !!python/name:material.extensions.emoji.to_svg
  - pymdownx.highlight:
      anchor_linenums: true
  - pymdownx.inlinehilite
  - pymdownx.snippets
  - pymdownx.superfences
  - pymdownx.tabbed:
      alternate_style: true
  - pymdownx.tasklist:
      custom_checkbox: true
  - tables
  - toc:
      permalink: true

plugins:
  - search
  - mkdocstrings:
      handlers:
        python:
          paths: [src] # Onde buscar pelo código fonte
```

options:

docstring_style: google

show_source: true

show_submodules: true

nav:

- Início: index.md
- Guia do Usuário:
 - user-guide/getting-started.md
 - user-guide/advanced-usage.md
- Documentação da API:
 - Visão Geral: api/index.md
 - Application: api/application.md
 - Domain: api/domain.md
 - Infrastructure: api/infrastructure.md
 - Interface: api/interface.md
 - Shared: api/shared.md
- Desenvolvimento:
 - development/contributing.md
 - development/architecture.md
- Changelog: changelog.md

extra:

social:

- icon: fontawesome/brands/github
link: <https://github.com/seu-usuario>
- icon: fontawesome/brands/twitter
link: <https://twitter.com/seu-usuario>
- icon: fontawesome/brands/linkedin
link: <https://linkedin.com/in/seu-usuario>

copyright: Copyright © 2025 Pereira Dev

18. Arquivo: models.py

```
# ./src/dev_platform/infrastructure/database/models.py
# -*- coding: utf-8 -*-
"""
Este módulo define os modelos de banco de dados usando SQLAlchemy,
permitindo a persistência de entidades do domínio em um banco de dados relacional.
"""

from sqlalchemy import Column, Integer, String, UniqueConstraint
from sqlalchemy.orm import declarative_base

Base = declarative_base()

class UserModel(Base):
    __tablename__ = "users"
    __table_args__ = (UniqueConstraint('email', name='uq_users_email'),)
    id = Column(Integer, primary_key=True, index=True)
    name = Column(String(100), nullable=False)
    email = Column(String(100), nullable=False, unique=True)

    def __repr__(self):
        return f"<UserModel(id={self.id}, name='{self.name}', email='{self.email}')>"
```

19. Arquivo: mypy.ini

```
[mypy]
ignore_missing_imports = True
plugins = pydantic.mypy,sqlalchemy.ext.mypy.plugin
python_version = 3.11
```

20. Arquivo: poetry.toml

```
[virtualenvs]  
in-project = true  
create = true
```

21. Arquivo: ports.py

```
# ./src/dev_platform/application/user/ports.py
# -*- coding: utf-8 -*-
"""
Este módulo define as interfaces para os repositórios e serviços relacionados à entidade User.
"""

from abc import ABC, abstractmethod

from dev_platform.domain.user.interfaces import IUserRepository

class UnitOfWork(ABC):
    """
    Interface for the Unit of Work pattern.
    """
    users: IUserRepository

    @abstractmethod
    async def __aenter__(self):
        """Enter the context of the unit of work."""
        raise NotImplementedError

    @abstractmethod
    async def __aexit__(self, exc_type, exc_val, exc_tb):
        """Exit the context of the unit of work."""
        raise NotImplementedError
```

22. Arquivo: pyproject.toml

```
# ./pyproject.toml
[project]
name = "dev-platform"
version = "0.1.0"
description = ""
authors = [
    {name = "Sergio Pereira",email = "sergiopereira.br@hotmail.com"}
]
readme = "README.md"
requires-python = ">=3.11,<4.0"
dependencies = [
    "pymysql (>=1.1.1,<2.0.0)",
    "aiomysql (>=0.2.0,<0.3.0)",
    "typer (>=0.15.4,<0.16.0)",
    "pydantic[email] (>=2.11.7,<3.0.0)",
    "loguru (>=0.7.3,<0.8.0)",
    "uuid (>=1.30,<2.0)",
    "alembic (>=1.16.1,<2.0.0)",
    "sqlalchemy (>=2.0.41,<3.0.0)",
    "python-dotenv (>=1.1.0,<2.0.0)",
    "pandas (>=2.3.0,<3.0.0)",
    "cryptography (>=45.0.4,<46.0.0)",
    "pydantic-settings (>=2.9.1,<3.0.0)",
]

[tool.poetry]
packages = [{include = "dev_platform", from = "src"}]

[tool.poetry.group.dev.dependencies]
black = "^23.0"
flake8 = "^6.0"
taskipy = "^1.14.1"
reportlab = "^4.4.1"
chardet = "^5.2.0"
pypandoc = "^1.15"
mypy = "^1.16.0"
pytest-asyncio = "^1.0.0"
types-reportlab = "^4.4.1.20250602"
matplotlib = "^3.10.3"
numpy = "^2.3.0"
python-dateutil = "^2.9.0.post0"

[tool.poetry.group.docs.dependencies]
mkdocs = "^1.6.1"
pymdown-extensions = "^10.15"
mkdocstrings = "^0.29.1"
mkdocstrings-python = "^1.16.10"
sphinx = "^6.0"
sphinx-rtd-theme = "^1.0"
mkdocs-material = "^9.6.14"

[tool.poetry.group.test.dependencies]
pytest = "^8.3.5"

[build-system]
requires = ["poetry-core>=2.0.0,<3.0.0"]
build-backend = "poetry.core.masonry.api"

# Configuração opcional para definir o script principal
[tool.poetry.scripts]
dev-platform = "dev_platform.main:main" # Isso cria um comando executável
docs-serve = "mkdocs:serve" # Iniciar servidor de documentação local
```

```
docs-build = "mkdocs:build" # Construir a documentação estática
```

```
[tool.taskipy.tasks]
```

```
# Definição de tarefas para o projeto
```

```
format = "poetry run black ./src ./tests ./scripts"
```

```
lint = "poetry run flake8 ./src ./tests ./scripts"
```

```
test = "poetry run pytest --cov=dev_platform --cov-report=html --cov-report=term ./tests"
```

```
test-async = "poetry run pytest-asyncio --cov=dev_platform --cov-report=html --cov-report=term ./tests"
```

```
"
```

```
docs-serve = "poetry run mkdocs serve" # Iniciar servidor de documentação local
```

```
docs-build = "poetry run mkdocs build" # Construir a documentação estática
```

```
docs-serve-live = "poetry run mkdocs serve --livereload" # Iniciar servidor de documentação com recarregamento automático
```

```
docs-serve-live-port = "poetry run mkdocs serve --livereload --dev-addr 0.0.0.0:8000"
```

```
# Tarefas para gerenciamento de usuários - testes e desenvolvimento
```

```
list = "poetry run python ./src/dev_platform/main.py user list-users"
```

```
list_debug = "poetry run python -m debugpy --listen 5678 --wait-for-client ./src/dev_platform/main.py user list-users"
```

```
create = "poetry run python ./src/dev_platform/main.py user create-user --name \"Sergio da Silva Pereira\" --email sergio.hertz@gmail.com"
```

```
delete = "poetry run python ./src/dev_platform/main.py user delete-user --email 'sergiopereira.br@hotmail.com'"
```

```
update = "poetry run python ./src/dev_platform/main.py user update-user --email 'sergiopereira.br@hotmail.com' --name 'Sergio P.'"
```

```
# Tarefas gerenciamento de usuários
```

```
list_users = "poetry run python ./src/dev_platform/main.py user list-users"
```

```
create_user = "poetry run python ./src/dev_platform/main.py user create-user --name 'Sergio da Silva Pereira' --email 'sergio.hertz@gmail.com'"
```

```
delete_user = "poetry run python ./src/dev_platform/main.py user delete-user --email 'sergiopereira.br@hotmail.com'"
```

```
update_user = "poetry run python ./src/dev_platform/main.py user update-user --email 'sergiopereira.br@hotmail.com' --name 'Sergio P.'"
```

```
# Tarefas para gerenciamento de projetos - testes e desenvolvimento
```

```
list_projects = "poetry run python ./src/dev_platform/main.py project list-projects"
```

```
create_project = "poetry run python ./src/dev_platform/main.py project create-project --name 'Projeto Teste' --description 'Descrição do Projeto Teste'"
```

```
delete_project = "poetry run python ./src/dev_platform/main.py project delete-project --name 'Projeto Teste'"
```

```
update_project = "poetry run python ./src/dev_platform/main.py project update-project --name 'Projeto Teste' --description 'Descrição Atualizada do Projeto Teste'"
```

```
# Tarefas para manipulação de arquivos e geração de relatórios
```

```
clean = "pwsh -Command ./scripts/tools/del_folders.ps1 './src' '__pycache__'"
```

```
stack = "pwsh -Command ./scripts/tools/stack_files.ps1 './src' 'py'"
```

```
stack2pdf = "poetry run python ./scripts/tools/pdf_generator.py ./src/stake_file --title \"Arquivos do Projeto DEV Platform\" -v"
```

```
compilado = "poetry run python ./scripts/tools/pdf_generator.py ./src/stake_file --title \"Arquivos do Projeto DEV Platform\" -v"
```

```
# Tarefas para conversão de arquivos Markdown para Word
```

```
# (melhor executar no WSL Ubuntu 20.04)
```

```
analise_md2word = "poetry run python ./scripts/tools/md_to_word.py ./docs/análise_26.md"
```

```
onboarding_md2word = "poetry run python ./scripts/tools/md_to_word.py ./docs/onboarding_26.md"
```

```
# Tarefas para análise de logs
```

```
analisa_log = "poetry run python ./scripts/tools/log_analyzer_mvp.py --log_folder_path ./scripts/requisitos_log"
```

```
[tool.mypy]
```

```
python_version = "3.11"
```

```
plugins = "pydantic.mypy"
```

23. Arquivo: README.md

<Arquivo vazio ou sem conteúdo legível>

24. Arquivo: repositories.py

```
# ./src/dev_platform/infrastructure/database/repositories.py
# -*- coding: utf-8 -*-
"""
This module implements the SQLAlchemy repository for the User entity,
providing methods for CRUD operations and queries.
"""

from typing import List, Optional
from sqlalchemy.ext.asyncio import AsyncSession
from sqlalchemy.future import select
from sqlalchemy import delete, func
from sqlalchemy.exc import SQLAlchemyError, IntegrityError
from dev_platform.domain.user.interfaces import IUserRepository
from dev_platform.domain.user.entities import User
from dev_platform.domain.user.value_objects import UserName, Email
from dev_platform.application.ports.logger import ILogger
from dev_platform.domain.exceptions import (
    DataIntegrityException,
    DatabaseException,
)
from dev_platform.domain.user.user_exceptions import (
    UserAlreadyExistsException,
    UserNotFoundException
)
from dev_platform.infrastructure.database.models import UserModel
from dev_platform.application.ports.logger import ILogger

class SQLUserRepository(IUserRepository):
    """SQLAlchemy implementation of the IUserRepository interface."""
    def __init__(self, session: AsyncSession, logger: ILogger):
        self._session = session
        self._logger = logger

    def _convert_to_domain_user(self, db_user: UserModel) -> User:
        """Convert database model to domain entity."""
        try:
            return User(
                id=db_user.id, name=UserName(db_user.name), email=Email(db_user.email)
            )
        except ValueError as e:
            self._logger.error(
                f"Data conversion error for user {db_user.id}: {str(e)}"
            )
            # This should not happen if database constraints are properly set
            raise DatabaseException(
                operation="data_conversion",
                reason=f"Invalid data in database: {str(e)}",
                original_exception=e,
            )

    async def add(self, user: User) -> User:
        """Adds a new user to the database."""
        if user.id is not None:
            raise DatabaseException(operation="add", reason="Cannot add a user that already has an ID.")
        try:
            db_user = UserModel(name=user.name.value, email=user.email.value)
            self._session.add(db_user)
            await self._session.flush()
            return user.with_id(db_user.id) # Retorna nova instância com ID
        except IntegrityError as e:
            # Transforma erro de violação de constraint em exceção de domínio
            raise UserAlreadyExistsException(user.email.value) from e
```



```

except SQLAlchemyError as e:
    raise DatabaseException(operation="add", reason=str(e), original_exception=e)

async def update(self, user: User) -> User:
    """Updates an existing user in the database."""
    if user.id is None:
        raise UserNotFoundException("None")
    try:
        # O find_by_id já foi feito no caso de uso, podemos ir direto para o merge ou update.
        # A forma mais segura é buscar e atualizar para garantir que o registro existe.
        db_user = await self._session.get(UserModel, user.id)
        if not db_user:
            raise UserNotFoundException(str(user.id))

        await self._session.merge(
            UserModel(id=user.id, name=user.name.value, email=user.email.value)
        )
        await self._session.flush()
        return user
    except SQLAlchemyError as e:
        raise DatabaseException(operation="update", reason=str(e), original_exception=e)

async def find_by_email(self, email: str) -> Optional[User]:
    """Find a user by email address."""
    try:
        result = await self._session.execute(
            select(UserModel).where(UserModel.email == email)
        )
        db_user = result.scalars().first()

        if db_user:
            return self._convert_to_domain_user(db_user)
        return None
    except SQLAlchemyError as e:
        self._logger.error(
            "SQLAlchemy error in find_by_email",
            extra={
                "operation": "find_by_email",
                "email": email,
                "error": str(e)
            }
        )
        RepositoryExceptionHandler.handle_sqlalchemy_error(operation="find_by_email", error=e, email=email)

async def find_all(self) -> List[User]:
    """Find all users in the database."""
    try:
        result = await self._session.execute(select(UserModel))
        db_users = result.scalars().all()

        return [self._convert_to_domain_user(db_user) for db_user in db_users]
    except SQLAlchemyError as e:
        self._logger.error(
            "SQLAlchemy error in find_all_users",
            extra={
                "operation": "find_all_users",
                "error": str(e)
            }
        )
        RepositoryExceptionHandler.handle_sqlalchemy_error(operation="find_all_users", error=e)
    return []

```

```

async def find_by_id(self, user_id: int) -> Optional[User]:
    """Find a user by ID."""
    try:
        result = await self._session.execute(
            select(UserModel).where(UserModel.id == user_id)
        )
        db_user = result.scalars().first()

        if db_user:
            return self._convert_to_domain_user(db_user)
        return None

    except SQLAlchemyError as e:
        self._logger.error(
            "SQLAlchemy error in find_by_id",
            extra={
                "operation": "find_by_id",
                "user_id": user_id,
                "error": str(e)
            }
        )
        RepositoryExceptionHandler.handle_sqlalchemy_error(
            operation="find_by_id", error=e, user_id=user_id
        )
    return None

async def find_by_ids(self, user_ids: List[int]) -> List[User]:
    """Find users by a list of IDs."""
    try:
        result = await self._session.execute(
            select(UserModel).where(UserModel.id.in_(user_ids))
        )
        return [self._convert_to_domain_user(u) for u in result.scalars().all()]
    except SQLAlchemyError as e:
        self._logger.error(
            "SQLAlchemy error in find_by_ids",
            extra={
                "operation": "find_by_ids",
                "user_ids": user_ids,
                "error": str(e)
            }
        )
        RepositoryExceptionHandler.handle_sqlalchemy_error(operation="find_by_ids", error=e, user_ids=
user_ids)
    return []

async def delete(self, user_id: int) -> bool:
    """Delete a user by ID."""
    try:
        # First check if user exists
        existing_user = await self.find_by_id(user_id)
        if not existing_user:
            raise UserNotFoundException(str(user_id))

        # Perform deletion
        result = await self._session.execute(
            delete(UserModel).where(UserModel.id == user_id)
        )

        success = result.rowcount > 0
        if success:
            await self._session.flush()

        return success

    except UserNotFoundException:

```

```

        # Re-raise domain exceptions as-is
        raise
    except SQLAlchemyError as e:
        self._logger.error(
            "SQLAlchemy error in delete_user",
            extra={
                "operation": "delete_user",
                "user_id": user_id,
                "error": str(e)
            }
        )
        RepositoryExceptionHandler.handle_sqlalchemy_error(
            operation="delete_user", error=e, user_id=user_id
        )
    return False

    async def find_by_name_contains(self, name_part: str) -> List[User]:
        """Find users whose name contains the given string."""
        try:
            result = await self._session.execute(
                select(UserModel).where(UserModel.name.contains(name_part))
            )
            db_users = result.scalars().all()

            return [self._convert_to_domain_user(db_user) for db_user in db_users]

        except SQLAlchemyError as e:
            self._logger.error(
                "SQLAlchemy error in find_by_name_contains",
                extra={
                    "operation": "find_by_name_contains",
                    "name_part": name_part,
                    "error": str(e)
                }
            )
            RepositoryExceptionHandler.handle_sqlalchemy_error(
                operation="find_by_name_contains", error=e, name_part=name_part
            )
        return []

    async def count(self) -> int: # Nota
        """Count total number of users."""
        try:
            result = await self._session.execute(select(func.count(UserModel.id)))
            count = result.scalar()
            return count if count is not None else 0

        except SQLAlchemyError as e:
            self._logger.error(
                "SQLAlchemy error in count_users",
                extra={
                    "operation": "count_users",
                    "error": str(e)
                }
            )
            RepositoryExceptionHandler.handle_sqlalchemy_error(operation="count_users", error=e)

    return 0 # Nota de teste: Retornar 0 se a contagem falhar, o que é mais intuitivo do que retornar None.

class RepositoryExceptionHandler:
    """Utility class for handling repository exceptions consistently."""

    @staticmethod
    def handle_sqlalchemy_error(operation: str, error: SQLAlchemyError, **context):

```

```

"""Handle SQLAlchemy specific errors."""
if isinstance(error, IntegrityError):
    #Acessa o erro original do driver DBAPI
    dbapi_exception = error.orig

    # Checa se o driver fornece o nome da constraint (ex: psycopg2, mysql-connector)
    # A forma exata pode variar um pouco entre drivers, mas o princípio é o mesmo.
    # Para PyMySQL/aiomysql, a análise do erro pode ser necessária.
    # No entanto, a forma mais robusta é checar o código de erro do MySQL.
    # Erro 1062 do MySQL é para entrada duplicada.

    # Exemplo para MySQL (código de erro 1062)
    if hasattr(dbapi_exception, 'errno') and dbapi_exception.errno == 1062:
        error_msg = str(dbapi_exception)
        if "'uq_users_email'" in error_msg: # Checa o nome da constraint
            email = context.get("email", "unknown")
            raise UserAlreadyExistsException(email) from error
        else:
            # Outra violação de unicidade
            raise DataIntegrityException(
                constraint_name="unknown", # ou extrair o nome da constraint da msg
                details=error_msg, original_exception=error
            ) from error

    context_str = ", ".join([f"{k}={v}" for k, v in context.items()])
    error_msg = f"{operation} failed"
    if context_str:
        error_msg += f" ({context_str})"

    raise DatabaseException(
        operation=operation, reason=str(error), original_exception=error
    )

@staticmethod
def handle_generic_error(operation: str, error: Exception, **context):
    """Handle generic errors."""
    context_str = ", ".join([f"{k}={v}" for k, v in context.items()])
    error_msg = f"{operation} failed"
    if context_str:
        error_msg += f" ({context_str})"

    raise DatabaseException(
        operation=operation, reason=str(error), original_exception=error
    )

```

25. Arquivo: services.py

```
# ./src/dev_platform/domain/user/services.py
# -*- coding: utf-8 -*-
"""
Este módulo define os serviços de domínio relacionados à entidade User.
Os serviços são responsáveis por validações complexas, regras de negócio e operações
com usuários.
"""

from abc import ABC, abstractmethod
from typing import List, Dict, Optional, Set
import re
from datetime import datetime
from dev_platform.domain.user.interfaces import IUserRepository
from dev_platform.domain.user.entities import User
from dev_platform.domain.exceptions import DatabaseException
from dev_platform.domain.user.user_exceptions import (
    UserValidationException,
    UserAlreadyExistsException,
    UserNotFoundException,
    EmailDomainNotAllowedException,
)

# --- Regras de validação devem ser extraídas para um módulo próprio (ex: validation_rules.py) ---
# Aqui mantemos apenas a interface base para uso no domínio.

class ValidationRule(ABC):
    """Base class for validation rules."""

    @abstractmethod
    async def validate(self, user: User) -> Optional[str]:
        """
        Validate user according to this rule.
        Returns None if valid, error message if invalid.
        """
        pass

    @property
    @abstractmethod
    def rule_name(self) -> str:
        pass

class UserValidatorService: # Novo serviço focado em validação de regras de negócio
    def __init__(self, validation_rules: List):
        self._validation_rules = validation_rules

    async def validate(self, user: User) -> None: # Método principal de validação
        validation_errors = {}
        for rule in self._validation_rules:
            error_message = await rule.validate(user)
            if error_message:
                validation_errors[rule.rule_name] = error_message
        if validation_errors:
            raise UserValidationException(validation_errors)

    # Métodos como validate_user_update, validate_user_creation_constraints,
    # validate_business_domain_rules (se forem puramente de validação de regras)
    # seriam movidos para cá ou para regras de validação específicas.
    # Métodos de gerenciamento de regras (add/remove/summary) seriam movidos para ValidationRulePr
    ovider ou um Registry.

# --- Serviço de domínio focado apenas na lógica de negócio ---
# UserDomainService seria renomeado ou refatorado para UserValidatorService
class UserUniquenessService:
```

```
"""Service focused on uniqueness validation."""
```

```
def __init__(self, user_repository: IUserRepository):  
    self._repository = user_repository
```

```
async def ensure_email_is_unique(  
    self, email: str, exclude_user_id: Optional[int] = None  
) -> None:  
    existing_user = await self._repository.find_by_email(email)  
    if existing_user and (  
        exclude_user_id is None or existing_user.id != exclude_user_id  
    ):  
        raise UserAlreadyExistsException(email)
```

```
class UserDomainService:
```

```
    """  
    Serviço de domínio para validações complexas de domínio de usuário e regras de negócio.  
    Recebe explicitamente as regras de validação a serem aplicadas.  
    """
```

```
def __init__(self, validation_rules: List[ValidationRule]):  
    self._validation_rules = validation_rules
```

```
def add_validation_rule(self, rule: ValidationRule):  
    """Add a custom validation rule."""  
    self._validation_rules.append(rule)
```

```
def remove_validation_rule(self, rule_name: str):  
    """Remove a validation rule by name."""  
    self._validation_rules = [  
        rule for rule in self._validation_rules if rule.rule_name != rule_name  
    ]
```

```
async def validate_business_rules(self, user: User) -> None:
```

```
    """  
    Valida todas as regras de negócio para um usuário.  
    Levanta UserValidationException se alguma regra falhar.  
    """
```

```
    validation_errors = {}
```

```
    # Run all validation rules  
    for rule in self._validation_rules:  
        error_message = await rule.validate(user)  
        if error_message:  
            validation_errors[rule.rule_name] = error_message  
    if validation_errors:  
        raise UserValidationException(validation_errors)
```

```
async def validate_user_update(self, user_id: int, updated_user: User) -> None:
```

```
    """  
    Validate user update, checking uniqueness only if email changed.  
    """
```

```
    validation_errors = {}
```

```
    # Run validation rules  
    for rule in self._validation_rules:  
        try:  
            error_message = await rule.validate(updated_user)  
            if error_message:  
                validation_errors[rule.rule_name] = error_message  
        except Exception as e:  
            validation_errors[rule.rule_name] = f"Validation rule failed: {str(e)}"
```

```
    if validation_errors:  
        raise UserValidationException(validation_errors)
```

```

def get_validation_summary(self) -> Dict[str, str]:
    """Get summary of all active validation rules."""
    return {
        rule.rule_name: rule.__class__.__doc__ or "No description available"
        for rule in self._validation_rules
    }

async def validate_user_creation_constraints(self, user: User) -> None:
    """
    Validate constraints specific to user creation.
    (Exemplo: limite de usuários, regras de negócio específicas)
    """
    validation_errors = {}

    try:
        current_count = await self._repository.count()
        if current_count >= 10000: # Exemplo de limite
            validation_errors["system_limit"] = "Maximum number of users reached"
    except Exception as e:
        validation_errors["system_check"] = f"Unable to verify system constraints: {str(e)}"

    if validation_errors:
        raise UserValidationException(validation_errors)

async def validate_business_domain_rules(
    self, user: User, domain_whitelist: Optional[List[str]] = None
) -> None:
    """
    Validate business-specific domain rules.
    """
    if domain_whitelist:
        email_domain = user.email.value.split("@")[1].lower()
        if email_domain not in [d.lower() for d in domain_whitelist]:
            raise EmailDomainNotAllowedException(
                user.email.value, email_domain, domain_whitelist
            )

class UserAnalyticsService:
    """Service for user analytics and reporting."""

    def __init__(self, user_repository: IUserRepository):
        self._repository = user_repository

    async def get_user_statistics(self) -> Dict[str, int]:
        """Get basic user statistics."""
        try:
            total_users = await self._repository.count()
            return {
                "total_users": total_users,
            }
        except Exception as e:
            raise DatabaseException("count", str(e), e)

    async def find_users_by_domain(self, domain: str) -> List[User]:
        """Find all users with emails from a specific domain."""
        try:
            all_users = await self._repository.find_all()
            return [
                user
                for user in all_users
                if user.email.value.split("@")[1].lower() == domain.lower()
            ]
        except Exception as e:
            raise DatabaseException("find_all", str(e), e)

```

26. Arquivo: session.py

```
# ./src/dev_platform/infrastructure/database/session.py
# -*- coding: utf-8 -*-
"""
Este módulo define o gerenciador de sessões de banco de dados,
centralizando a criação e gerenciamento de sessões síncronas e assíncronas usando SQLAlchemy.
Ele permite o uso de context managers para garantir o commit automático e rollback em caso de exceção
s.
"""
```

```
from contextlib import asynccontextmanager
from typing import AsyncGenerator, Optional
from sqlalchemy import create_engine
from sqlalchemy.orm import sessionmaker, Session
from sqlalchemy.engine import Engine
from sqlalchemy.ext.asyncio import create_async_engine, AsyncEngine, AsyncSession, async_sessionmaker
from dev_platform.infrastructure.config import ConfigurationFacade
```

```
class DatabaseSessionManager:
```

```
    """
    Gerenciador centralizado de sessões de banco de dados.
```

```
    ⚠ Commit automático no context manager
```

O método `get_async_session` realiza automaticamente um `await session.commit()` ao sair do bloco `async with`.

Se ocorrer uma exceção dentro do bloco, será feito `await session.rollback()` antes de propagar o erro.

Atenção:

Esse comportamento pode surpreender desenvolvedores que esperam controlar o commit manualmente.

Se você precisa de controle total sobre commit/rollback, utilize a factory de sessão diretamente.

Exemplo de uso:

`async with db_manager.get_async_session() as session:`

Todas as operações dentro deste bloco serão automaticamente commitadas ao final,

ou rollback em caso de exceção.

...

Resumo:

O commit é automático ao sair do bloco.

O rollback é automático em caso de exceção.

Para controle manual, crie a sessão diretamente via factory."""

```
def __init__(self, config: Optional[ConfigurationFacade] = None):
    self._async_engine: Optional[AsyncEngine] = None
    self._sync_engine: Optional[Engine] = None
    self._async_session_factory: Optional[async_sessionmaker] = None
    self._sync_session_factory: Optional[sessionmaker] = None
    self.config = config or ConfigurationFacade()
    self._initialize_engines()
```

```
def _initialize_engines(self):
    """Inicializa os engines síncronos e assíncronos."""
    # Configurações do pool
    pool_config = {
        "pool_size": int(self.config.get("database_pool_size", 5)),
        "max_overflow": int(self.config.get("database_max_overflow", 10)),
        "pool_pre_ping": self.config.get("database_pool_pre_ping", True),
    }
```

```
    # Engine assíncrono
```

```
    async_url = self.config.get("database_url")
```



```

self._async_engine = create_async_engine(
    async_url,
    echo=self.config.get("database_echo", False),
    **pool_config
)

# Session factory assíncrona
self._async_session_factory = async_sessionmaker(
    bind = self._async_engine, class_=AsyncSession, expire_on_commit=False
)

# Engine síncrono (se necessário para migrações ou outras operações)
if not async_url.startswith(
    "sqlite+aiosqlite"
): # SQLite não precisa de engine síncrono separado
    sync_url = self.config.get("database_url")
    self._sync_engine = create_engine(
        sync_url, echo = self.config.get("database_echo", False), **pool_config
    )

    self._sync_session_factory = sessionmaker(
        bind = self._sync_engine, autocommit=False, autoflush=False
    )

def get_async_session_factory(self):
    return self._async_session_factory

@asynccontextmanager
async def get_async_session(self) -> AsyncGenerator[AsyncSession, None]:
    """
    Fornece uma sessão de banco de dados assíncrona.
    O controle de commit e rollback é delegado ao chamador (ex: Unit of
    Work).
    """
    if self._async_session_factory is None:
        raise RuntimeError("Async session factory is not initialized")
    session = self._async_session_factory()
    try:
        yield session
    finally:
        await session.close()

def get_sync_session(self) -> Session:
    """Obtém uma sessão síncrona (para migrações, etc)."""
    if not self._sync_session_factory:
        raise RuntimeError("Sync session not available for this database type")
    return self._sync_session_factory()

async def close_async_engine(self):
    """Fecha o engine assíncrono."""
    if self._async_engine:
        await self._async_engine.dispose()

def close_sync_engine(self):
    """Fecha o engine síncrono."""
    if self._sync_engine:
        self._sync_engine.dispose()

@property
def async_engine(self) -> AsyncEngine:
    """Propriedade para acessar o engine assíncrono."""
    return self._async_engine

@property
def sync_engine(self) -> Engine:

```

```
        """Propriedade para acessar o engine síncrono."""
        return self._sync_engine
```

```
# Instância global do gerenciador de sessões
db_manager = DatabaseSessionManager(config=ConfigurationFacade())
```

```
# Funções de conveniência para compatibilidade
```

```
async def get_async_session():
```

```
    """Função de conveniência para obter sessão assíncrona.
```

```
    Garante que o gerenciador de sessões assíncronas esteja inicializado
    e faz await session.commit() automaticamente ao sair do bloco."""
```

```
    if db_manager._async_session_factory is None:
```

```
        raise RuntimeError("Async session factory is not initialized")
```

```
    async with db_manager.get_async_session() as session:
```

```
        yield session
```

```
def get_sync_session():
```

```
    """Função de conveniência para obter sessão síncrona."""
```

```
    return db_manager.get_sync_session()
```

```
# Aliases para compatibilidade com código existente
```

```
if db_manager._async_session_factory:
```

```
    AsyncSessionLocal = db_manager._async_session_factory
```

```
if db_manager._sync_session_factory:
```

```
    SessionLocal = db_manager._sync_session_factory
```

27. Arquivo: settings.json

```
{
  "python.terminal.activateEnvironment": true,
  "python.defaultInterpreterPath": "${workspaceFolder}/dev_platform/.venv/Scripts/python.exe",
  "python.analysis.extraPaths": [
    "${workspaceFolder}/dev_platform/src"
  ]
}
```

28. Arquivo: structured_logger.py

```
# src/dev_platform/infrastructure/logging/structured_logger.py
# -*- coding: utf-8 -*-
"""
Este módulo implementa um logger estruturado usando Loguru,
com suporte a níveis dinâmicos e correlação de logs.
Ele permite a configuração de handlers para diferentes níveis de log
e formatos, incluindo JSON para fácil integração com sistemas de monitoramento.
"""

from typing import Optional
import os
from uuid import uuid4
from loguru import logger
from dev_platform.infrastructure.config import ConfigurationFacade
from dev_platform.application.ports.logger import ILogger

class StructuredLogger(ILogger):
    """Logger estruturado usando Loguru com suporte a níveis dinâmicos e correlação de logs."""

    def __init__(self, name: str = "DEV Platform", config: Optional[ConfigurationFacade] = None):
        self._name = name
        self._config = config
        self._configure_logger()

    def _configure_logger(self):
        """Configura o logger com base no ambiente e adiciona handlers."""
        # Remover handlers padrão do Loguru
        logger.remove()

        # Obter nível de log com base no ambiente
        # Usa a configuração injetada
        environment = self._config.get("environment", "production") if self._config else "production"
        log_level = self._config.get("logging_level", "INFO").upper() if self._config else "INFO"
        log_levels = {"development": "DEBUG", "test": "DEBUG", "production": "INFO"}
        default_level = log_levels.get(environment, "INFO")
        final_level = (
            log_level
            if log_level in ["DEBUG", "INFO", "WARNING", "ERROR", "CRITICAL"]
            else default_level
        )

        # Configurar handler para console (JSON, todos os níveis)
        logger.add(
            sink="sys.stdout",
            level=final_level,
            format="{time:YYYY-MM-DD HH:mm:ss.SSS} | {level} | {message} | {extra}",
            serialize=True, # Formato JSON
        )

        # Configurar handler para arquivo (apenas ERROR, com rotação)
        if not os.path.exists("logs"):
            os.makedirs("logs")
        logger.add(
            sink=f"logs/{self._name}_{{time:YYYY-MM-DD}}.log",
            level="ERROR",
            rotation="10 MB",
            retention="5 days",
            compression="zip",
            enqueue=True, # Assíncrono
            serialize=True, # <-- Adicione esta linha para JSON
        )
```

```

def set_correlation_id(self, correlation_id: Optional[str] = None):
    """Define um ID de correlação para rastreamento."""
    self._logger = logger.bind(correlation_id=correlation_id or str(uuid4()))

def info(self, message: str, **kwargs):
    """Registra uma mensagem de nível INFO."""
    (self._logger if hasattr(self, "_logger") else logger).bind(**kwargs).info(message)

def error(self, message: str, **kwargs):
    """Registra uma mensagem de nível ERROR."""
    (self._logger if hasattr(self, "_logger") else logger).bind(**kwargs).error(message)

def warning(self, message: str, **kwargs):
    """Registra uma mensagem de nível WARNING."""
    (self._logger if hasattr(self, "_logger") else logger).bind(**kwargs).warning(message)

def debug(self, message: str, **kwargs):
    """Registra uma mensagem de nível DEBUG."""
    (self._logger if hasattr(self, "_logger") else logger).bind(**kwargs).debug(message)

def critical(self, message: str, **kwargs):
    """Registra uma mensagem de nível CRITICAL."""
    (self._logger if hasattr(self, "_logger") else logger).bind(**kwargs).critical(message)

# NOVO MÉTODO PARA SHUTDOWN GRACIOSO DO LOGGER
@staticmethod
def shutdown():
    """
    Garante que todas as mensagens enfileiradas pelo Loguru sejam processadas
    e que os handlers sejam removidos. Isso é crucial para limpar recursos
    assíncronos do logger antes que o loop de eventos feche.
    """
    logger.shutdown() # Processa todas as mensagens enfileiradas
    logger.remove() # Remove todos os handlers para evitar vazamentos de recursos

```

29. Arquivo: tree.txt

```
./tree.txt
# Directory structure for the DEV Platform project
# Esse arquivo contém a estrutura de diretórios do projeto DEV Platform
# e é utilizado para gerar a documentação do projeto.
# A estrutura é gerada automaticamente pelo comando `tree` no terminal.
```

```
dev_platform
| .env
| .env.development
| .env.production
| .env.test
| .gitignore
| alembic.ini
| config.production.json
| mkdocs.yml
| mypy.ini
| mypy.txt
| poetry.lock
| poetry.toml
| pyproject.toml
| README.md
| sys.stdout
| tree.txt
| tree3.txt
|
+---.venv
| | pyenv.cfg
| | Scripts
| |   activate
| |   activate.bat
| |   deactivate.bat
| |   pip.exe
| |   python.exe
| |   pythonw.exe
| |   ...
| |
| \---Lib
| |   site-packages
| |   ...
| |
| \---...
+---docs
|
+---logs
|   DEV Platform_2025-06-01.log
|   DEV Platform_2025-06-02.log
|   DEV Platform_2025-06-03.log
|   DEV Platform_2025-06-05.log
|   DEV Platform_2025-06-06.log
|   DEV Platform_2025-06-07.log
|   DEV Platform_2025-06-08.log
|   DEV Platform_2025-06-09.log
|   DEV Platform_2025-06-11.log
|   DEV Platform_2025-06-12.log
|   DEV Platform_2025-06-13.log
|   DEV Platform_2025-06-14.log
|   DEV Platform_2025-06-15.log
|   DEV Platform_2025-06-17.log
|
+---migrations
| | env.py
| | README
| | script.py.mako
```

```

| |
| | \---versions
| |
+---scripts
| | +---log
| | | amostra_log.pdf
| | | conceito.pdf
| |
| | \---tools
| | | cria_pastas.ps1
| | | cria_pastas_arquivo.ps1
| | | del_folderes.ps1
| | | log_analyzer_mvp.py
| | | md_to_word.py
| | | pdf_generator.py
| | | stack_files.ps1
| | | tree.ps1
| |
+---tests
| | test_db.py
| |
\---src
| | +---dev_platform
| | | main.py
| | |
| | | +---application
| | | | +---ports
| | | | | logger.py
| | | |
| | | | \---__pycache__
| | | | | logger.cpython-311.pyc
| | | |
| | | | \---user
| | | | | dtos.py
| | | | | mappers.py
| | | | | ports.py
| | | | | use_cases.py
| | | |
| | | | \---__pycache__
| | | | | dtos.cpython-311.pyc
| | | | | ports.cpython-311.pyc
| | | | | use_cases.cpython-311.pyc
| | |
| | | +---domain
| | | | exceptions.py
| | | | validation_rules.py
| | |
| | | +---user
| | | | entities.py
| | | | interfaces.py
| | | | services.py
| | | | user_exceptions.py
| | | | value_objects.py
| | |
| | | \---__pycache__
| | | | entities.cpython-311.pyc
| | | | interfaces.cpython-311.pyc
| | | | services.cpython-311.pyc
| | | | user_exceptions.cpython-311.pyc
| | | | value_objects.cpython-311.pyc
| | |
| | | \---__pycache__
| | | | exceptions.cpython-311.pyc
| | | | validation_rules.cpython-311.pyc
| | |
| | +---infrastructure

```

```

| | composition_root.py
| | config.py
| |
| | +---database
| | | models.py
| | | repositories.py
| | | session.py
| | | unit_of_work.py
| | |
| | \---__pycache__
| | | models.cpython-311.pyc
| | | repositories.cpython-311.pyc
| | | session.cpython-311.pyc
| | | unit_of_work.cpython-311.pyc
| |
| | +---logging
| | | structured_logger.py
| | |
| | \---__pycache__
| | | structured_logger.cpython-311.pyc
| |
| | \---__pycache__
| | | composition_root.cpython-311.pyc
| | | config.cpython-311.pyc
| |
| | \---interface
| | | \---cli
| | | | user_commands.py
| | | |
| | | \---__pycache__
| | | | user_commands.cpython-311.pyc
| |
| | \---stake_file
| | .env
| | .env.development
| | .env.production
| | .env.test
| | .gitignore
| | alembic.ini
| | composition_root.py
| | config.production.json
| | config.py
| | dtos.py
| | entities.py
| | env.py
| | exceptions.py
| | interfaces.py
| | logger.py
| | main.py
| | mappers.py
| | mkdocs.yml
| | models.py
| | mypy.ini
| | poetry.toml
| | ports.py
| | pyproject.toml
| | README.md
| | repositories.py
| | services.py
| | session.py
| | settings.json
| | structured_logger.py
| | tree.txt
| | unit_of_work.py
| | user_commands.py
| | user_exceptions.py

```


use_cases.py
validation_rules.py
value_objects.py

30. Arquivo: unit_of_work.py

```
# ./src/dev_platform/infrastructure/database/unit_of_work.py
# -*- coding: utf-8 -*-
"""
Este módulo implementa o padrão Unit of Work para gerenciar transações de banco de dados
e repositórios de usuários usando SQLAlchemy.
"""

from typing import Optional
from contextlib import AbstractAsyncContextManager

from sqlalchemy.ext.asyncio import AsyncSession

from dev_platform.infrastructure.config import ConfigurationFacade
from dev_platform.application.ports.logger import ILogger
from dev_platform.infrastructure.logging.structured_logger import StructuredLogger
from dev_platform.infrastructure.database.session import db_manager

from dev_platform.application.user.ports import UnitOfWork
from dev_platform.domain.user.interfaces import IUserRepository # Importa a interface
from dev_platform.infrastructure.database.repositories import SQLUserRepository # Ainda precisa da impl
ementação concreta para instanciar

class SQLUnitOfWork(UnitOfWork):
    def __init__(self, logger: Optional[ILogger] = None, user_repository: Optional[IUserRepository] = None):
        # Recebe IUserRepository
        self.session_context: Optional[AbstractAsyncContextManager[AsyncSession]] = None # Gerenciador
        de contexto para a sessão assíncrona
        self.logger: ILogger = logger or StructuredLogger(CONFIG__=ConfigurationFacade())
        self.user_repository: Optional[IUserRepository] = user_repository # Atribui o repositório injetado
        self.session: Optional[AsyncSession] = None

    @property
    def user_repository(self) -> IUserRepository:
        if self._user_repository is None:
            self.logger.error("O repositório de usuários não foi inicializado no contexto do Unit of Work.")
            raise RuntimeError("O repositório de usuários não foi inicializado no contexto do Unit of Work.")
        )
        return self._user_repository

    async def __aenter__(self):
        # Usar o gerenciador de sessões
        self.session_context = db_manager.get_async_session()
        self.session = await self.session_context.__aenter__()
        # Garante que o repositório injetado use a sessão correta
        if self._user_repository:
            # Se o repositório foi injetado, atualiza sua sessão interna
            if hasattr(self._user_repository, '_session'): # Assumindo que a implementação concreta tem _sessi
on
                self._user_repository._session = self.session
            else:
                # Alternativa: o repositório é uma factory que recebe a sessão
                pass # Lógica mais complexa para factories, fora do escopo inicial
        else:
            # Fallback: se não injetado, cria um SQLUserRepository aqui, mas isso é menos ideal
            self._user_repository = SQLUserRepository(self.session, logger=self.logger) # Causa Raiz: Instanci
ação concreta
        return self

    async def __aexit__(self, exc_type, exc_val, exc_tb):
        # Delega TODA a lógica de saída para o gerenciador de sessão.
        # O gerenciador já cuida do commit, rollback e fechamento.
        if self.session_context:
```

```
        await self._session_context.__aexit__(exc_type, exc_val, exc_tb)
    # Limpa as referências
    self._session = None
    self._user_repository = None
    self._session_context = None

    async def commit(self):
        await self._session.commit()

    async def rollback(self):
        await self._session.rollback()
```

31. Arquivo: use_cases.py

```
# ./src/dev_platform/application/user/use_cases.py
# -*- coding: utf-8 -*-
"""
Este módulo define os casos de uso para a entidade User,
encapsulando a lógica de negócios e as interações com os repositórios.
"""

from typing import List

from dev_platform.application.user.dtos import UserCreateDTO, UserUpdateDTO, UserDTO
from dev_platform.domain.user.entities import User
from dev_platform.application.ports.logger import ILogger
from dev_platform.application.user.ports import UnitOfWork
from dev_platform.domain.exceptions import DatabaseException
from dev_platform.domain.user.user_exceptions import (
    UserValidationException,
    UserAlreadyExistsException,
    UserNotFoundException
)
from dev_platform.domain.user.services import UserDomainService, UserUniquenessService, UserValidatorService

# Função Helper para conversão do entity para DTO
def user_to_dto(user: User) -> UserDTO:
    return UserDTO(
        id=str(user.id),
        name=user.name.value if hasattr(user.name, "value") else user.name,
        email=user.email.value if hasattr(user.email, "value") else user.email,
    )

class BaseUseCase:
    """Base class for all use cases, providing access to Unit of Work and logger."""
    def __init__(self, uow: UnitOfWork, logger: ILogger):
        self._uow = uow
        self._logger = logger

class CreateUserUseCase(BaseUseCase):
    """Caso de uso para criar um novo usuário."""
    def __init__(
        self,
        uow: UnitOfWork,
        logger: ILogger,
        user_validator: UserValidatorService,
        user_uniqueness_service: UserUniquenessService,
    ):
        super().__init__(uow, logger)
        self._user_validator = user_validator
        self._user_uniqueness_service = user_uniqueness_service

    async def execute(self, dto: UserCreateDTO) -> UserDTO:
        async with self._uow:
            self._logger.info("Iniciando criação de usuário", name=dto.name, email=dto.email)
            try:
                await self._user_uniqueness_service.ensure_email_is_unique(dto.email)
                user_to_create = User.create(name=dto.name, email=dto.email)
                await self._user_validator.validate(user_to_create)

                saved_user = await self._uow.user_repository.add(user_to_create)
                await self._uow.commit()
                self._logger.info(
                    "Usuário criado com sucesso",
                    user_id=saved_user.id,
                    name=saved_user.name.value if hasattr(saved_user.name, "value") else saved_user.name,
                )
            except DatabaseException as e:
                self._logger.error(f"Erro ao criar usuário: {e}")
                raise e
```

```

        email=saved_user.email.value if hasattr(saved_user.email, "value") else saved_user.email,
    )
    return user_to_dto(saved_user)
except UserAlreadyExistsException as e:
    await self._uow.rollback()
    self._logger.warning(
        "Validação de domínio tentou criar usuário duplicado", email=dto.email
    )
    raise
except UserValidationException as e:
    await self._uow.rollback()
    self._logger.warning(
        "Validação de regras de negócio falhou", validation_errors=e.validation_errors
    )
    raise
except Exception as e:
    await self._uow.rollback()
    self._logger.error(
        "Erro durante a criação do usuário",
        error=str(e),
    )
    raise

class ListUsersUseCase(BaseUseCase):
    """Use case for listing all users."""
    async def execute(self) -> List[UserDTO]:
        async with self._uow:
            try:
                self._logger.info("Starting user listing")
                users = await self._uow.user_repository.find_all()
                self._logger.info("Users retrieved successfully", count=len(users))
                return [user_to_dto(user) for user in users]
            except UserNotFoundException:
                self._logger.error("User not found")
                raise

class UpdateUserUseCase(BaseUseCase):
    """Use case for updating an existing user."""
    def __init__(
        self,
        uow: UnitOfWork,
        logger: ILogger,
        domain_service: UserDomainService,
    ):
        super().__init__(uow, logger)
        self._domain_service = domain_service

    async def execute(self, user_id: int, dto: UserUpdateDTO) -> UserDTO:
        async with self._uow:
            self._logger.set_correlation_id()
            self._logger.info(
                "Starting user update", user_id=user_id, update_data=dto.model_dump()
            )
            try:
                existing_user = await self._uow.user_repository.find_by_id(user_id)
                if not existing_user:
                    self._logger.error("User not found for update", user_id=user_id)
                    raise UserNotFoundException(str(user_id))

                new_name = dto.name if dto.name is not None else existing_user.name.value
                new_email = dto.email if dto.email is not None else existing_user.email.value
                updated_user = existing_user.update_details(new_name, new_email)

                saved_user = await self._uow.user_repository.update(updated_user)
                await self._uow.commit()
                saved_user_dto = user_to_dto(saved_user)

```

```

        self._logger.info(
            "User updated successfully",
            user_id=saved_user_dto.id,
            name=saved_user_dto.name,
            email=saved_user_dto.email,
        )
        return saved_user_dto
    except (UserValidationException, UserNotFoundException) as e:
        if isinstance(e, UserValidationException):
            self._logger.error(
                "User update validation failed",
                user_id=user_id,
                validation_errors=e.validation_errors,
            )
        else:
            self._logger.error("User not found for update", user_id=user_id)
            raise
    except Exception as e:
        self._logger.error(
            "Domain error during user update",
            user_id=user_id,
            error=str(e),
        )
        raise

class GetUserUseCase(BaseUseCase):
    """Use case for retrieving a user by ID."""
    async def execute(self, user_id: int) -> UserDTO:
        async with self._uow:
            try:
                self._logger.info("Getting user", user_id=user_id)
                user = await self._uow.user_repository.find_by_id(user_id)
                if not user:
                    self._logger.error("User not found", user_id=user_id)
                    raise UserNotFoundException(str(user_id))
                self._logger.info("User retrieved successfully", user_id=user_id)
                return user_to_dto(user)
            except UserNotFoundException:
                self._logger.error("User not found", user_id=user_id)
                raise

class DeleteUserUseCase(BaseUseCase):
    """Use case for deleting a user by ID."""
    async def execute(self, user_id: int) -> bool:
        async with self._uow:
            try:
                self._logger.info("Starting user deletion", user_id=user_id)
                existing_user = await self._uow.user_repository.find_by_id(user_id)
                if not existing_user:
                    self._logger.error("User not found for deletion", user_id=user_id)
                    raise UserNotFoundException(str(user_id))
                success = await self._uow.user_repository.delete(user_id)
                if success:
                    await self._uow.commit()
                    self._logger.info("User deleted successfully", user_id=user_id)
                else:
                    self._logger.warning("User deletion failed", user_id=user_id)
                return success
            except UserNotFoundException:
                self._logger.error("User not found for deletion", user_id=user_id)
                raise
    except Exception as e:
        self._logger.error(
            "Domain error during user deletion",
            user_id=user_id,
            error=str(e),
        )

```

)
raise

32. Arquivo: user_commands.py

```
# ./src/dev_platform/client/cli/user_commands.py
# -*- coding: utf-8 -*-
"""
CLI para gerenciamento de usuários no DEV Platform.
Este módulo define comandos CLI para criar, listar, atualizar, obter e excluir usuários.
"""

import asyncio
import click
import sys, os
from typing import Optional, List
from dev_platform.infrastructure.config import ConfigurationFacade
from dev_platform.application.user.dtos import UserCreatedDTO, UserUpdateDTO, UserDTO
from dev_platform.infrastructure.composition_root import CompositionRoot
from dev_platform.infrastructure.database.unit_of_work import SQLUnitOfWork
from dev_platform.application.ports.logger import ILogger
from dev_platform.infrastructure.logging.structured_logger import StructuredLogger
from dev_platform.domain.exceptions import ConfigurationException

# Logger global para uso em run_async
_LOGGER: ILogger = StructuredLogger()

def run_async(coro) -> None:
    """
    Executa uma corrotina de forma segura em qualquer ambiente.
    Usa run_until_complete em CLI puro e asyncio.run se possível.
    Lança erro amigável se já houver um loop rodando (ex: Jupyter).
    """
    try:
        loop = asyncio.get_event_loop()
        if loop.is_running():
            raise RuntimeError(
                "Já existe um loop de eventos rodando. "
                "Execute este comando em um terminal/CLI puro."
            )
        return loop.run_until_complete(coro)
    except RuntimeError as re:
        _LOGGER.critical(
            f"Runtime error occurred: {re}",
            exception=str(re)
        )
    sys.exit(1) # Encerra o CLI com erro

class UserCommands:
    def __init__(self, composition_root: CompositionRoot, logger: ILogger):
        self._composition_root = composition_root
        self._logger = logger
        # try:
        #     self._composition_root = CompositionRoot(
        #         environment=config.get("ENVIRONMENT", "production"),
        #         logger=self._logger,
        #         validation_rule_provider=ValidationRuleProvider()
        #     )
        # except ConfigurationException as ce:
        #     self._logger.critical(
        #         f"Configuration error during initialization: {ce}",
        #         exception=str(ce)
        #     )
        #     raise
        # except Exception as e:
        #     self._logger.critical(
        #         f"Unexpected error during initialization: {e}",
```



```

#     exception=str(e)
# )
#     raise ConfigurationException(
#         config_key="COMPOSITION_ROOT",
#         reason=f"Unexpected error: {e}"
#     )

```

async def create_user_async(self, name: str, email: str) -> str:

```

"""
Cria um novo usuário.
"""
try:
    async with SQLUnitOfWork(self._logger) as uow:
        repo = uow.user_repository
        use_case = self._composition_root.create_user_use_case(uow, repo)
        dto: UserCreateDTO = UserCreateDTO(name=name, email=email)
        user: UserDTO = await use_case.execute(dto)
        name_val = getattr(user.name, "value", user.name)
        email_val = getattr(user.email, "value", user.email)
        return f"User created successfully: ID {user.id}, Name: {name_val}, Email: {email_val}"
except ValueError as e:
    self._logger.warning(f"Validation error: {e}")
    return f"Error: Validation Error: {e}"
except ConfigurationException as ce:
    self._logger.error(f"Configuration error: {ce}", exception=str(ce))
    return f"Error: Configuration Error: {ce}"
except Exception as e:
    self._logger.error(f"Error creating user: {e}", exception=str(e))
    return f"Error: {e}"

```

async def list_users_async(self) -> List[str]:

```

"""
Lista todos os usuários.
"""
try:
    async with SQLUnitOfWork(self._logger) as uow:
        repo = uow.user_repository
        use_case = self._composition_root.list_users_use_case(uow, repo)
        users: List[UserDTO] = await use_case.execute()
        if not users:
            return ["No users found"]

        result: List[str] = []
        for user in users:
            name_val = getattr(user.name, "value", user.name)
            email_val = getattr(user.email, "value", user.email)
            result.append(
                f"ID: {user.id}, Name: {name_val}, Email: {email_val}"
            )
        return result
except ConfigurationException as ce:
    self._logger.error(f"Configuration error: {ce}", exception=str(ce))
    return [f"Error: Configuration Error: {ce}"]
except Exception as e:
    self._logger.error(f"Error listing users: {e}", exception=str(e))
    return [f"Error: {e}"]

```

async def update_user_async(
 self, user_id: int, name: Optional[str] = None, email: Optional[str] = None
) -> str:

```

"""
Atualiza um usuário existente.
"""
try:
    # A CLI agora apenas passa os dados, sem lógica de preenchimento
    async with SQLUnitOfWork(self._logger) as uow:

```

```

        repo = uow.user_repository
        # O DTO agora aceita valores nulos
        update_dto = UserUpdateDTO(name=name, email=email)
        update_use_case = self._composition_root.update_user_use_case(uow, repo)
        # O caso de uso agora tem toda a responsabilidade
        updated_user = await update_use_case.execute(user_id=user_id, dto=update_dto)
        return f"User {user_id} updated successfully: Name: {updated_user.name}, Email: {updated_user.email}"
    except Exception as e:
        self._logger.error(f"Error updating user: {e}", exception=str(e))
        return f"Error: {e}"

    async def get_user_async(self, user_id: int) -> str:
        """
        Obtém um usuário pelo ID.
        """
        try:
            async with SQLUnitOfWork(self._logger) as uow:
                repo = uow.user_repository
                use_case = self._composition_root.get_user_use_case(uow, repo)
                user_entity = await use_case.execute(user_id=user_id)
                if not user_entity:
                    return f"User with ID {user_id} not found."
                name_val = getattr(user_entity.name, "value", user_entity.name)
                email_val = getattr(user_entity.email, "value", user_entity.email)
                return f"User found: ID {user_entity.id}, Name: {name_val}, Email: {email_val}"
        except ConfigurationException as ce:
            self._logger.error(f"Configuration error: {ce}", exception=str(ce))
            return f"Error: Configuration Error: {ce}"
        except Exception as e:
            self._logger.error(f"Error getting user: {e}", exception=str(e))
            return f"Error: {e}"

    async def delete_user_async(self, user_id: int) -> str:
        """
        Exclui um usuário pelo ID.
        """
        try:
            async with SQLUnitOfWork(self._logger) as uow:
                repo = uow.user_repository
                use_case = self._composition_root.delete_user_use_case(uow, repo)
                success: bool = await use_case.execute(user_id=user_id)
                if success:
                    return f"User {user_id} deleted successfully."
                else:
                    return f"User {user_id} could not be deleted (not found or other issue)."
        except ConfigurationException as ce:
            self._logger.error(f"Configuration error: {ce}", exception=str(ce))
            return f"Error: Configuration Error: {ce}"
        except Exception as e:
            self._logger.error(f"Error deleting user: {e}", exception=str(e))
            return f"Error: {e}"

# Instância única para a sessão da CLI
logger = StructuredLogger()
#composition_root = CompositionRoot(environment="production", logger=logger) # Criada uma única vez

# Ponto de entrada cria as dependências de infraestrutura
def get_dependencies():
    config = ConfigurationFacade(environment=os.getenv("ENVIRONMENT",
"production"), logger=logger)
    composition_root = CompositionRoot(config=config, logger=logger)
    return composition_root

```

```

@click.group()
def cli():
    pass

@cli.command()
@click.option("--name", prompt="User name")
@click.option("--email", prompt="User email")
def create_user(name: str, email: str):
    """Create a new user."""
    composition_root = get_dependencies()
    commands: UserCommands = UserCommands(composition_root, logger)

    async def _run_create():
        result: str = await commands.create_user_async(name, email)
        click.echo(result)

    return run_async(_run_create())

@cli.command()
def list_users():
    """List all users."""
    composition_root = get_dependencies()
    commands: UserCommands = UserCommands(composition_root, logger)

    async def _run_list():
        results: List[str] = await commands.list_users_async()
        for line in results:
            click.echo(line)

    return run_async(_run_list())

@cli.command()
@click.option("--user-id", type=int, prompt="User ID to update")
@click.option(
    "--name",
    prompt="New user name (leave empty to keep current)",
    default="",
    show_default=False,
    help="Novo nome do usuário. Deixe em branco para manter o atual."
)
@click.option(
    "--email",
    prompt="New user email (leave empty to keep current)",
    default="",
    show_default=False,
    help="Novo e-mail do usuário. Deixe em branco para manter o atual."
)
def update_user(user_id: int, name: str, email: str):
    """Update an existing user."""
    composition_root = get_dependencies()
    commands: UserCommands = UserCommands(composition_root, logger)

    async def _run_update():
        result: str = await commands.update_user_async(
            user_id, name if name else None, email if email else None
        )
        click.echo(result)

    return run_async(_run_update())

@cli.command()
@click.option("--user-id", type=int, prompt="User ID to retrieve")
def get_user(user_id: int):
    """Get a user by ID."""
    composition_root = get_dependencies()
    commands: UserCommands = UserCommands(composition_root, logger)

```

```
async def _run_get():
    result: str = await commands.get_user_async(user_id)
    click.echo(result)

return run_async(_run_get())

@cli.command()
@click.option("--user-id", type=int, prompt="User ID to delete")
def delete_user(user_id: int):
    """Delete a user by ID."""
    composition_root = get_dependencies()
    commands: UserCommands = UserCommands(composition_root, logger)

    async def _run_delete():
        result: str = await commands.delete_user_async(user_id)
        click.echo(result)

    return run_async(_run_delete())
```

33. Arquivo: user_exceptions.py

```
# ./src/dev_platform/domain/user/user_exceptions.py
# -*- coding: utf-8 -*-
"""
Este módulo define exceções específicas para o domínio de usuários,
permitindo o tratamento de erros de forma consistente e informativa.
"""

from datetime import datetime
from typing import Optional, Dict, Any

class DomainException(Exception):
    """Base exception for all domain-related errors."""

    def __init__(
        self,
        message: str,
        error_code: Optional[str] = None,
        details: Optional[Dict[str, Any]] = None,
    ):
        self.message = message
        self.error_code = error_code or self.__class__.__name__
        self.details = details or {}
        self.timestamp = datetime.now()
        super().__init__(self.message)

    def to_dict(self) -> Dict[str, Any]:
        return {
            "error_code": self.error_code,
            "message": self.message,
            "details": self.details,
            "timestamp": self.timestamp.isoformat(),
        }

    def __str__(self):
        return f"[{self.error_code}] {self.message}"

class UserAlreadyExistsException(DomainException):
    """Raised when trying to create a user with an email that already exists."""

    def __init__(self, email: str):
        self.email = email
        super().__init__(
            message=f"User with email '{email}' already exists",
            error_code="USER_ALREADY_EXISTS",
            details={"email": email},
        )

class UserNotFoundException(DomainException):
    """Raised when a user cannot be found."""

    def __init__(self, identifier: str, identifier_type: str = "id"):
        self.identifier = identifier
        self.identifier_type = identifier_type
        super().__init__(
            message=f"User not found with {identifier_type}: {identifier}",
            error_code="USER_NOT_FOUND",
            details={"identifier": identifier, "identifier_type": identifier_type},
        )

class InvalidUserDataException(DomainException):
    """Raised when user data fails validation."""

    def __init__(self, field: str, value: Any, reason: str):
```

```

        self.field = field
        self.value = value
        self.reason = reason
        super().__init__(
            message=f"Invalid {field}: {reason}",
            error_code="INVALID_USER_DATA",
            details={"field": field, "value": str(value), "reason": reason},
        )

class UserValidationException(DomainException):
    """Raised when user business rules validation fails."""

    def __init__(self, validation_errors: Dict[str, str]):
        self.validation_errors = validation_errors
        errors_summary = ", ".join(
            [f"{field}: {error}" for field, error in validation_errors.items()]
        )
        super().__init__(
            message=f"User validation failed: {errors_summary}",
            error_code="USER_VALIDATION_FAILED",
            details={"validation_errors": validation_errors},
        )

class EmailDomainNotAllowedException(DomainException):
    """Raised when email domain is not in allowed list."""

    def __init__(self, email: str, domain: str, allowed_domains: list):
        self.email = email
        self.domain = domain
        self.allowed_domains = allowed_domains
        super().__init__(
            message=f"Email domain '{domain}' is not allowed. Allowed domains: {' '.join(allowed_domains)}",
            error_code="EMAIL_DOMAIN_NOT_ALLOWED",
            details={
                "email": email,
                "domain": domain,
                "allowed_domains": allowed_domains,
            },
        )

class UserOperationException(DomainException):
    """Raised when a user operation fails."""

    def __init__(self, operation: str, user_id: int, reason: str):
        self.operation = operation
        self.user_id = user_id
        self.reason = reason
        super().__init__(
            message=f"Failed to {operation} user {user_id}: {reason}",
            error_code="USER_OPERATION_FAILED",
            details={"operation": operation, "user_id": user_id, "reason": reason},
        )

# Compatibility aliases (deprecated, use specific exceptions above)
class DomainError(DomainException):
    """Exception for domain-related errors. DEPRECATED: Use DomainException instead."""

    def __init__(self, message: str):
        import warnings

        warnings.warn(
            "DomainError is deprecated. Use DomainException instead.",
            DeprecationWarning,
            stacklevel=2,
        )
        super().__init__(message)

```

```
class ValidationException(DomainException):
    """Exception for validation-related errors. DEPRECATED: Use UserValidationException instead."""

    def __init__(self, message: str):
        import warnings

        warnings.warn(
            "ValidationException is deprecated. Use UserValidationException instead.",
            DeprecationWarning,
            stacklevel=2,
        )
        super().__init__(message)
```

34. Arquivo: validation_rules.py

```
# ./src/dev_platform/domain/validation_rules.py
# -*- coding: utf-8 -*-
"""
Este módulo contém regras de validação para diversas operações no domínio de usuários.
"""

from abc import ABC, abstractmethod
from typing import Optional, List, Set
import re
from datetime import datetime
from dev_platform.domain.user.entities import User

class ValidationRule(ABC):
    """Base class for validation rules."""

    @abstractmethod
    async def validate(self, user: User) -> Optional[str]:
        """
        Validate user according to this rule.
        Returns None if valid, error message if invalid.
        """
        pass

    @property
    @abstractmethod
    def rule_name(self) -> str:
        pass

class EmailFormatAdvancedValidationRule(ValidationRule):
    """Advanced email format validation beyond basic regex."""

    def __init__(self):
        self.pattern = re.compile(
            r"^[a-zA-Z0-9]([a-zA-Z0-9._-]*[a-zA-Z0-9])?@[a-zA-Z0-9]([a-zA-Z0-9.-]*[a-zA-Z0-9])?\.[a-zA-Z]{2,}$"
        )

    async def validate(self, user: User) -> Optional[str]:
        email = user.email.value

        if not self.pattern.match(email):
            return "Email format is invalid"

        if ".." in email:
            return "Email cannot contain consecutive dots"

        if len(email) > 254:
            return "Email is too long (max 254 characters)"

        local_part, domain_part = email.split("@")

        if len(local_part) > 64:
            return "Email local part is too long (max 64 characters)"

        if len(domain_part) > 253:
            return "Email domain part is too long (max 253 characters)"

        return None

    @property
    def rule_name(self) -> str:
        return "email_format_advanced_validation"
```



```

class NameContentValidationRule(ValidationRule):
    """Validates name content and format."""

    def __init__(self, allowed_chars: Optional[Set[str]] = None):
        if allowed_chars is None:
            allowed_chars = set(
                "abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ -'àáâãäåæçèéêëìíîïðóôõö÷ùúûüýÿÀÁÂÃÄÅÈÉÊËÌÍÎÏÐÓÔÕÖ÷ÙÚÛÜÝ"
            )
        self.allowed_chars = allowed_chars

    async def validate(self, user: User) -> Optional[str]:
        name = user.name.value

        if name.strip() != name:
            return "Name cannot start or end with whitespace"

        if " " in name:
            return "Name cannot contain consecutive spaces"

        if any(char.isdigit() for char in name):
            return "Name cannot contain numbers"

        if not all(char in self.allowed_chars for char in name):
            invalid_chars = [char for char in name if char not in self.allowed_chars]
            return f"Name contains invalid characters: {' '.join(set(invalid_chars))}"

        words = name.split()
        if len(words) < 2:
            return "Name must contain at least first and last name"

        for word in words:
            if len(word) < 2:
                return "Each name part must be at least 2 characters long"

        return None

    @property
    def rule_name(self) -> str:
        return "name_content_validation"

```

```

class EmailDomainValidationRule(ValidationRule):
    """Validates that email domain is in allowed list."""

    def __init__(self, allowed_domains: List[str]):
        self.allowed_domains = set(domain.lower() for domain in allowed_domains)

    async def validate(self, user: User) -> Optional[str]:
        email_domain = user.email.value.split("@")[1].lower()
        if email_domain not in self.allowed_domains:
            return f"Email domain '{email_domain}' is not allowed. Allowed domains: {' '.join(self.allowed_domains)}"
        return None

    @property
    def rule_name(self) -> str:
        return "email_domain_validation"

```

```

class ForbiddenWordsValidationRule(ValidationRule):
    """
    Valida se o nome do usuário contém palavras específicas proibidas.
    """

```

```

def __init__(self, forbidden_words: List[str]):
    self.forbidden_words = [word.lower() for word in forbidden_words]

async def validate(self, user: User) -> Optional[str]:
    for word in self.forbidden_words:
        if word in user.name.value.lower():
            return f"Name contains forbidden word: {word}"
    return None

@property
def rule_name(self) -> str:
    return "forbidden_words_validation"

class NameProfanityValidationRule(ValidationRule):
    """
    Valida se o nome do usuário contém palavras ou termos ofensivos.
    """

    def __init__(self, forbidden_words: List[str]):
        self.forbidden_words = [word.lower() for word in forbidden_words]

    async def validate(self, user: User) -> Optional[str]:
        name_lower = user.name.value.lower()
        for word in self.forbidden_words:
            if word in name_lower:
                return f"Name contains forbidden word: {word}"
        return None

    @property
    def rule_name(self) -> str:
        return "name_profanity_validation"

class IDatetimeProvider(ABC):
    @abstractmethod
    def now(self) -> datetime:
        pass

class SystemDateTimeProvider(IDatetimeProvider):
    def now(self) -> datetime:
        return datetime.now()

class BusinessHoursValidationRule(ValidationRule):
    """
    Regra de validação para verificar se uma operação está dentro do horário comercial permitido.

    Verifica o dia da semana (apenas dias úteis) e as horas (9 AM - 5 PM).

    Exemplo de uso:
        rule = BusinessHoursValidationRule(business_hours_only=True) # Usa o provedor padrão
        rule_test = BusinessHoursValidationRule(business_hours_only=True, datetime_provider=MockDateTi
meProvider(specific_time))
    """

    def __init__(self, business_hours_only: bool = False, datetime_provider: IDatetimeProvider = None):
        """
        Inicializa a regra de validação de horário comercial.

        Args:
            business_hours_only (bool): Se True, a validação de horário comercial será aplicada.
                Caso contrário, a regra será ignorada.
            datetime_provider (IDatetimeProvider, optional): Provedor de data/hora para testabilidade.
                Usa SystemDateTimeProvider por padrão.
        """
        self.business_hours_only = business_hours_only

```

```
        self.datetime_provider = datetime_provider if datetime_provider is not None else SystemDateTimePr  
ovider()
```

```
    async def validate(self, user: User) -> Optional[str]:
```

```
        """
```

```
        Valida se o registro do usuário está dentro do horário comercial permitido.
```

```
        Args:
```

```
            user (User): O objeto do usuário a ser validado.
```

```
        Returns:
```

```
            Optional[str]: Uma mensagem de erro se a validação falhar, None caso contrário.
```

```
        """
```

```
        if not self.business_hours_only:
```

```
            return None
```

```
        now = self.datetime_provider.now() # Alteração: usando o provedor injetado
```

```
        if now.weekday() >= 5:
```

```
            return "User registration only allowed during business days"
```

```
        if now.hour < 9 or now.hour >= 17:
```

```
            return "User registration only allowed during business hours (9 AM - 5 PM)"
```

```
        return None
```

```
    @property
```

```
    def rule_name(self) -> str:
```

```
        return "business_hours_validation"
```

35. Arquivo: value_objects.py

```
# ./src/dev_platform/domain/user/value_objects.py
# -*- coding: utf-8 -*-
"""
Este módulo define os Value Objects para o domínio de usuários, incluindo Email e UserName.
"""

from dataclasses import dataclass, field
import re
from abc import ABC, abstractmethod
from typing import Any

# Definição da interface de especificação
class ISpecification(ABC):
    """Interface genérica para especificações de domínio."""
    @abstractmethod
    def is_satisfied_by(self, candidate: Any) -> bool:
        """
        Verifica se o candidato satisfaz a especificação.

        Args:
            candidate (Any): O objeto a ser verificado.

        Returns:
            bool: True se o candidato satisfaz a especificação, False caso contrário.
        """
        pass

# Implementação da especificação de formato de email
class EmailFormatSpecification(ISpecification):
    """Especificação para validar o formato de um endereço de email."""
    EMAIL_REGEX = r"^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$"

    def is_satisfied_by(self, email_value: str) -> bool:
        """
        Verifica se a string de email fornecida tem um formato válido.

        Args:
            email_value (str): O endereço de email a ser validado.

        Returns:
            bool: True se o formato do email for válido, False caso contrário.
        """
        return re.match(self.EMAIL_REGEX, email_value) is not None

@dataclass(frozen=True)
class Email:
    """
    Representa um Value Object para endereço de email.

    Garante que o endereço de email seja válido no momento da criação.

    Exemplo de uso:
        email = Email("test@example.com")
        email_invalid = Email("invalid-email")
    """
    value: str
    # Injetando a especificação como uma dependência, com valor padrão para conveniência
    _email_format_spec: ISpecification = field(default_factory=EmailFormatSpecification, init=False, repr=False, compare=False)

    def __post_init__(self):
```

```
"""
```

Método chamado após a inicialização para validar o formato do email.

Raises:

ValueError: Se o formato do email for inválido.

```
"""
```

```
if not self._email_format_spec.is_satisfied_by(self.value): # Alteração: Usando a especificação
    raise ValueError(f"Invalid email format: {self.value}")
```

```
@dataclass(frozen=True)
```

```
class UserName:
```

```
    value: str
```

```
    def __post_init__(self):
```

```
        value = self.value.strip()
```

```
        if not value or len(value) < 3:
```

```
            raise ValueError("Name must be at least 3 characters long")
```

```
        if len(self.value) > 100:
```

```
            raise ValueError("Name cannot exceed 100 characters")
```